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AudioTester Tool User Guide

Issue V1.0

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About This Document

Purpose

This document describes UNISOC Audio tuning tool AudioTester.

Intended Audience

This document is intended primarily for UNISOC Audio tuning engineers.

Symbol Conventions

The symbols that may be found in this guide are defined in the following table.

Symbol	Description
NOTE	Calls attention to important information, best practices and tips. NOTE is used to address information not related to personal injury, equipment damage, and environment deterioration.
CAUTION	Calls attention to error-prone operations. CAUTION is used to address information not related to personal injury, equipment damage, and environment deterioration.
WARNING	Calls attention to irreversible operations. WARNING is used to address information not related to personal injury and environment deterioration.

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Acronyms and Abbreviations

Acronym and Abbreviation	Full Name
ST	Side Tone
NS	Noise Suppression
ASP	Acoustic Shock Protection
SWB	Super Wide Band
SFR	Sending Frequency Response
RFR	Receiving Frequency Response

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Acronym and Abbreviation	Full Name
DUT	Device Under Test
NB	Narrow Band
WB	Wide Band
FIR	Finite Impulse Response
IIR	Infinite Impulse Response
VOIP	Voice Over Internet Protocol
NR	Noise Reduction
CVS	Clear Voice Suite
SRC	Sampling Rate Convertor

Change History

Issue	Date	Description
V1.0	2021-03-20	This issue is the first official release.

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Keywords

AudioTester, Audio tuning, Multimedia.

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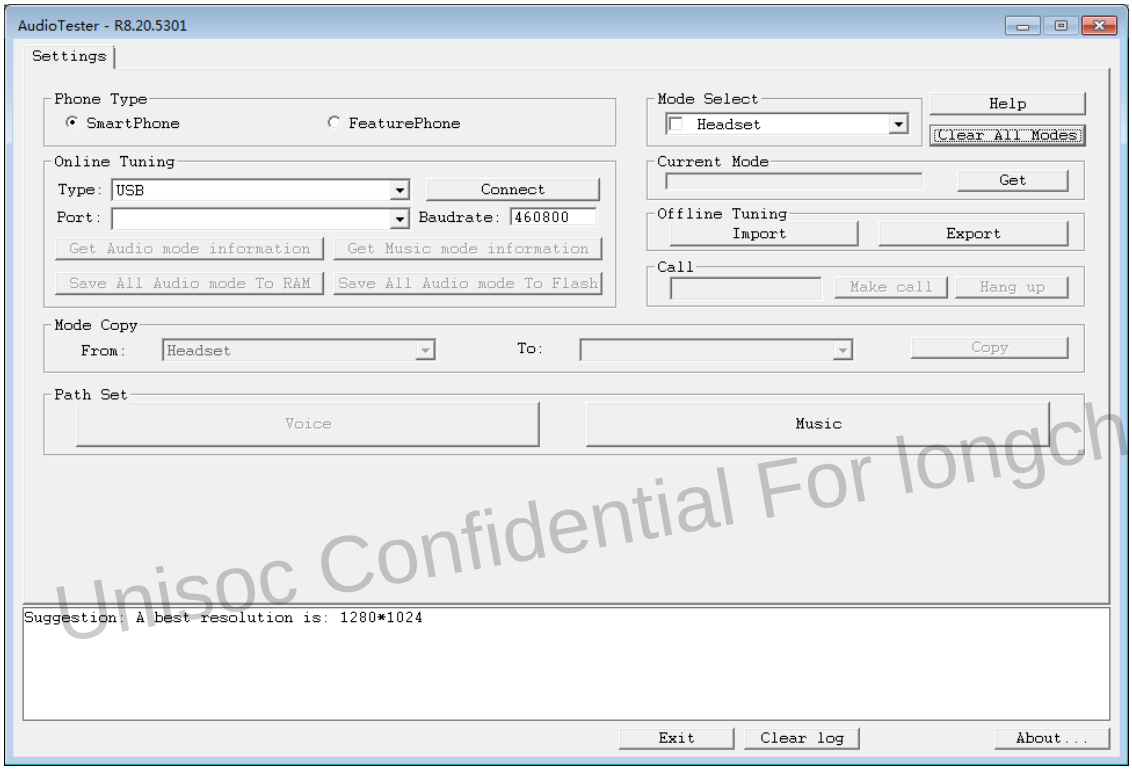
1 AudioTester

AudioTester is a UNISOC Audio parameters tuning tool, applying to SC9850K, SC9832E, SC9863A, SC7731E, SC6531 and other chip platforms. AudioTester interface is shown as **Figure 1-1**.

Figure 1-1 AudioTester interface



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1.1 Interface introduction

1.1.1 Phone Type

Select the right phone type before connecting phone with the tool.

Figure 1-2 Phone Type

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- FeaturePhone: for UNISOC feature phone, which means the OS is Mocor, such as SC6531.
- SmartPhone: for UNISOC smart phone, which means the OS is Android, KaiOS, YunOS, etc, such as SC9850K, SC9832E, SC9863A, SC7731.

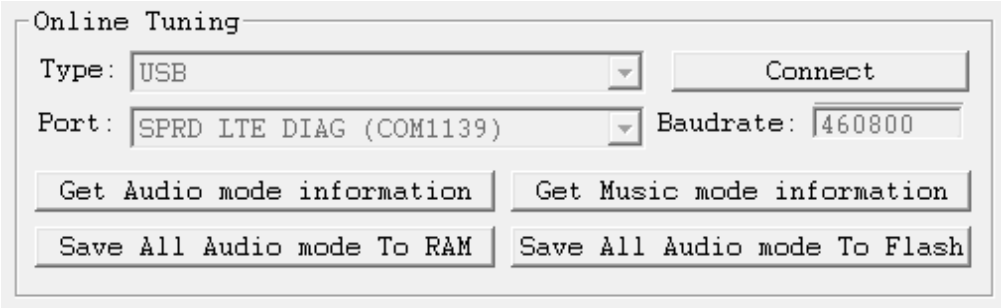
NOTE

Smart phone has adb command function, and feature phone does not has adb command.

1.1.2 Online Tuning

The setting of Online Tuning is shown as **Figure 1-3.**

Figure 1-3 Online Tuning



- Type: select USB
- Port: select port
  - Smart phone selects SPRD LTE DIAG (COMxxx)
  - Feature phone selects SPRD DIAG (COMxxx)
- Baudrate: select 460800
- Connect: click it after configuration to connect DUT with PC
- Get Audio mode information: read Audio related parameters from phone, including Voice Call, VOIP, Loopback, etc, and modem-related audio parameters.
- Get Music mode information: read multimedia related parameters from phone, including Music Play, FM, Record, etc, and AP/ARM related audio parameters.
- Save All Audio mode to RAM: write parameters of all audio modes currently in the tool to the RAM of DUT.
- Save All Audio mode to Flash: write parameters of all audio modes currently in the tool to the flash of DUT.

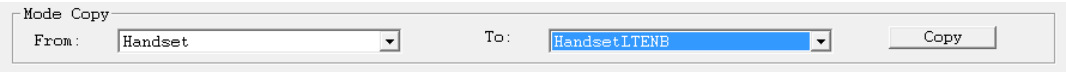
NOTE

- Write to RAM: if the power is off, parameters restore to the value before writing.
- Write to Flash: if the power is off, parameters retain the value after writing.

1.1.3 Mode Copy

This function is only available in Audio Mode. Select Get Audio mode information or Get Music mode information to enter the corresponding mode.

Figure 1-4 Mode Copy



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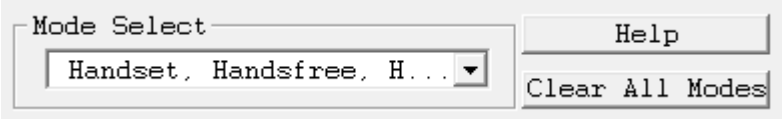
This function is only used to copy parameters in the same sampling rate and the same mode, for example, copying Handset parameters to HandsetLTENB, that is, copying GSM parameters in handset mode to LTE NB parameters in handset mode.

1.1.4 Mode Select

Select a corresponding mode under Mode Select to begin tuning after connecting the DUT to PC and selecting Get Audio mode information or Get Music mode information.

Figure 1-5 Mode Select



1.1.4.1 Audio Mode

The description of Mode Select options in Audio Mode is shown in **Table 1-1**.

Table 1-1 Description of Mode Select in Audio Mode

Mode	Description
Handset	receiver mode for GSM & WCDMA_NB
Handsfree	speaker mode for GSM & WCDMA_NB
Headset4P	4 pole earphone mode for GSM & WCDMA_NB
Headset3P	3 pole earphone mode for GSM & WCDMA_NB
HandsetTD	receiver mode for TD
HandsfreeTD	speaker mode for TD
Headset4PTD	4 pole earphone mode for TD
Headset3PTD	3 pole earphone mode for TD
HandsetWB	receiver mode for WCDMA_WB
HandsfreeWB	speaker mode for WCDMA_WB
Headset4PWB	4 pole earphone mode for WCDMA_WB
Headset3PWB	3 pole earphone mode for WCDMA_WB
BTHSD8K	8K BTHS mode (BT Headset no support NREC)
BTHSNRECD8K	8K BTHSNREC mode (BT Headset support NREC)
BTHSD16K	16K BTHS mode (BT Headset @16KHz sample rate, no support NRE)
BTHSNRECD16K	16K BTHSNREC mode (BT Headset @16KHz sample rate, support NREC)
HandsetVOIP1	receiver mode for VOIP Call

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Mode	Description
HandsfreeVOIP1	speaker mode for VOIP Call
Headset4PVOIP1	4 pole earphone mode for VOIP Call
Headset3PVOIP1	3 pole earphone mode for VOIP Call
BTHSD8K VOIP1	8K BTHS mode for VOIP Call
BTHSNRECD8K VOIP1	8K BTHSNREC mode for VOIP Call
BTHSD16K VOIP1	16K BTHS mode for VOIP Call
BTHSNRECD16K VOIP1	16K BTHSNREC mode for VOIP Call
HandsetLoopB	receiver for loopback
HandsfreeLoopB	speaker for loopback
Headset4PLoopB	4 pole earphone for loopback
Headset3PLoopB	3 pole earphone for loopback
HandsetLTENB	receiver mode for LTE_NB
HandsfreeLTENB	speaker mode for LTE_NB
Headset4PLTENB	4 pole earphone mode for LTE_NB
Headset3PLTENB	3 pole earphone mode for LTE_NB
HandsetLTEWB	receiver mode for LTE_WB
HandsfreeLTEWB	speaker mode for LTE_WB
Headset4PLTEWB	4 pole earphone mode for LTE_WB
Headset3PLTEWB	3 pole earphone mode for LTE_WB
BTHSD8KLTE	8K BTHS mode for LTE
BTHSNRECD8KLTE	8K BTHSNREC mode for LTE
BTHSD16KLTE	16K BTHS mode for LTE
BTHSNRECD16KLTE	16K BTHSNREC mode for LTE
HandsetVoWiNB	receiver mode for VOWIFI_NB
HandsfreeVoWiNB	speaker mode for VOWIFI_NB
Headset4PVoWiNB	4 pole earphone mode for VOWIFI_NB
Headset3PVoWiNB	3 pole earphone mode for VOWIFI_NB
HandsetVoWiWB	receiver mode for VOWIFI_WB
HandsfreeVoWiWB	speaker mode for VOWIFI_WB
Headset4PVoWiWB	4 pole earphone mode for VOWIFI_WB
Headset3PVoWiWB	3 pole earphone mode for VOWIFI_WB

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Mode	Description
BTHSD8KVoWi	8K BTHS mode for VOWIFI
BTHSNRECD8KVoWi	8K BTHSNREC mode for VOWIFI
BTHSD16KVoWi	16K BTHS mode for VOWIFI
BTHSNRECD16KVoWi	16K BTHSNREC mode for VOWIFI
HandsetSWB	receiver mode for LTE_SWB
HandsfreeSWB	speaker mode for LTE_SWB
Headset4PSWB	4 pole earphone mode for LTE_SWB
Headset3PSWB	3 pole earphone mode for LTE_SWB
HFWB_SAMP	External third-party Smart-AMP(IIS) parameter, 16KHz sample rate
HFTD_SAMP	External third-party Smart-AMP(IIS) parameter, TD
HFVOIP1_SAMP	External third-party Smart-AMP(IIS) parameter, VOIP
HFLoopB_SAMP	External third-party Smart-AMP(IIS) parameter, Loopback
HFVoWiWB_SAMP	External third-party Smart-AMP(IIS) parameter, VOWIFI
HFLTEWB_SAMP	External third-party Smart-AMP(IIS) parameter, LTE WB

NOTE

VoWifi calls VoLTE parameters.

- Help: show tool user guide.
- Clear All Modes: clear all modes that have been turned on.

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1.1.4.2 Music Mode

The description of Mode Select options in Music Mode is shown in **Table 1-2**.

Table 1-2 Description of Mode Select in Music Mode

Mode	Description
Handset	It applies to the receiver to play the sound during WeChat voice call.
Handsfree	It applies to the speaker to play the sound, including Music Play, FM and Audio Record.
Headset	It applies to the earphone to play the sound, including Music Play, FM and Audio Record.
Headfree	It applies to the earphone and the speaker to play the ringtone when the earphone is plugged in to the mobile phone.
ISCHandsfree	Smart-AMP (UNISOC feedforward) applies to the speaker to play the sound.
ISCHeadfree	Smart-AMP (UNISOC feedforward) applies to the earphone and the speaker to play the ringtone when the earphone is plugged in to the mobile phone.

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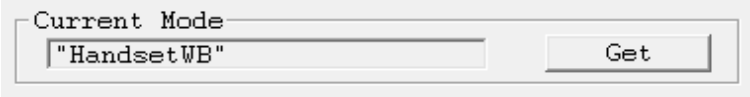
Mode	Description
Unprocess Record	It applies to the upper-layer applications to deliver sound recording that do not require audio algorithm processing. In this scene, turning off NR and ALC but turning on EQ to correct the frequency response curve problem caused by MIC and sound cavity is recommended.

1.1.5 Current Mode

Figure 1-6 Current Mode



Get: get the current mode of the phone, which is used after making a call.

1.1.6 Offline Tuning

Import or export NV files in offline tuning, interface is shown in **Figure 1-7**.

Figure 1-7 Offline Tuning

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- Import: import NV file.
- Export: export NV file.

NOTE

NV file is Audio parameter file generated by the tool.

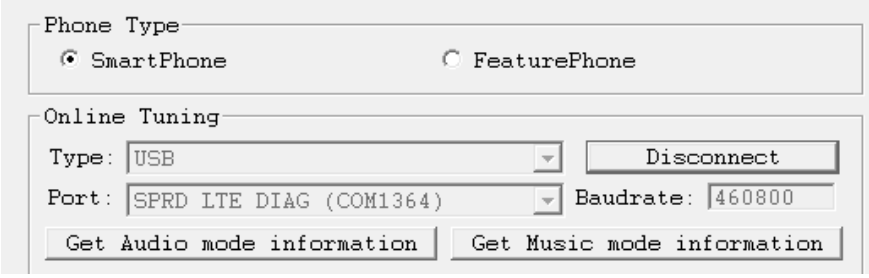
1.1.7 Path Set

Path Set includes the path selection under Audio mode and Music mode.

Audio Mode

Click **Get Audio mode information** to get Audio parameters.

Figure 1-8 Get Audio mode information

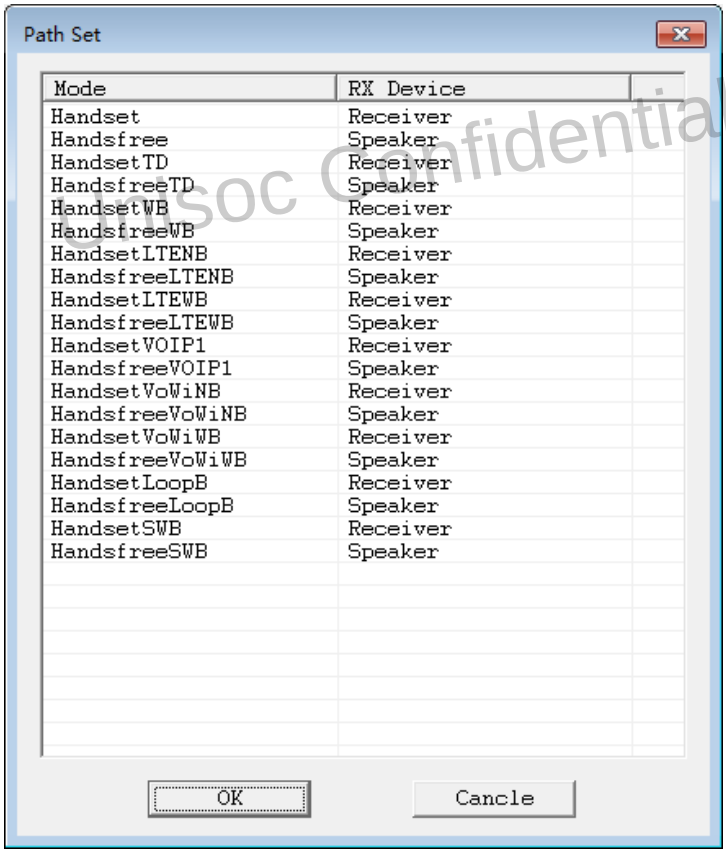


Voice in the Path Set is highlighted and Music is grayed out.

Figure 1-9 Path Set-Voice

Click **Voice**, and see **Figure 1-10**.

Figure 1-10 Path Set-Voice



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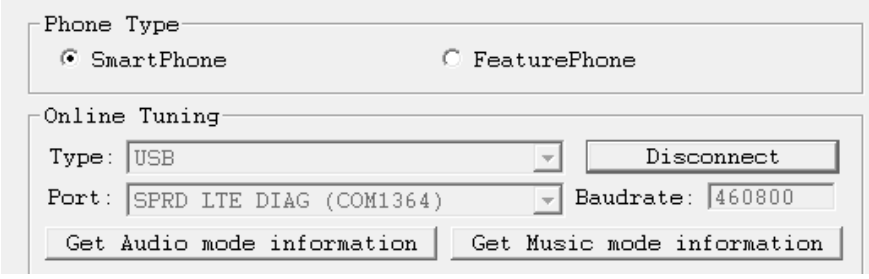
- Mode: see **1.1.4 Mode Select**

- RX Device: Codec output device
  - Receiver: receiver is working
  - Speaker: speaker is working

Music Mode

Click **Get Music mode information** to get multimedia parameters.

Figure 1-11 Get Music mode information



Music in the Path Set is highlighted and Voice is grayed out.

Figure 1-12 Pat Set–Music

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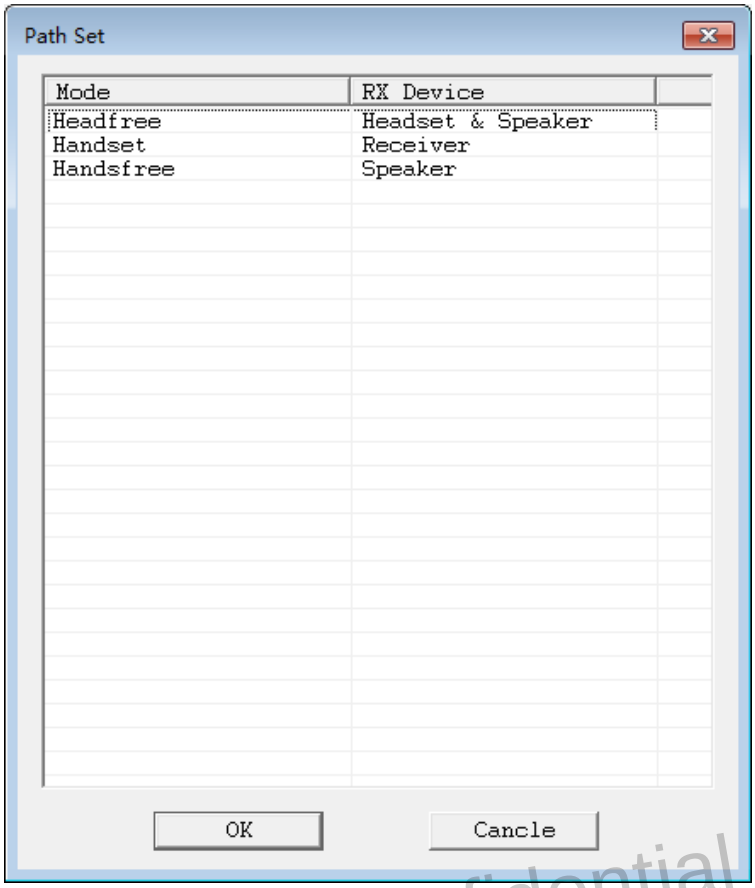
Click **Music**, and see **Figure 1-13**.

Figure 1-13 Path Set-Music



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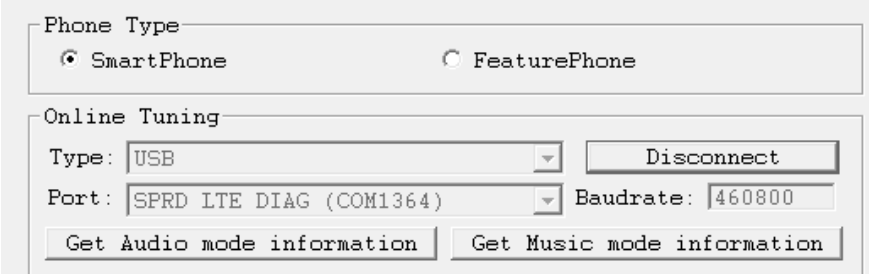
- Mode:
  - Headfree: handset& speaker mode
  - Handset: handset mode
  - Handsfree: speaker mode
- RX Device: Codec output device
  - Headset & Speaker: handset & speaker are working together
  - Receiver: receiver is working
  - Speaker: speaker is working

1.1.8 Call

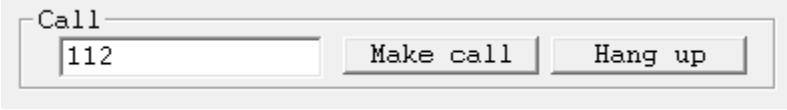
Click **Get Audio mode information** to get call parameters.

Figure 1-14 Get Audio mode information



Input telephone number in the edit box under Call.

Figure 1-15 Call



1.1.9 Log window

The tool's Log information is shown in **Figure 1-16**.

Figure 1-16 Log window

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1.2 Connect DUT with PC

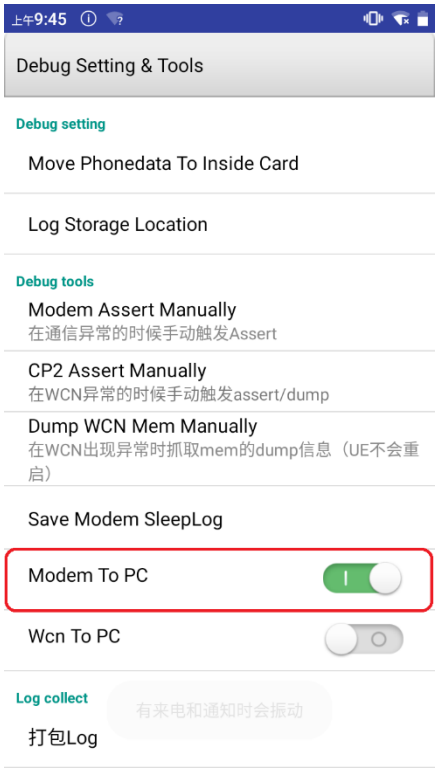
1.2.1 DUT settings

1. Download user debug version to the DUT.
2. Input ***#*#83781#*#*** in the dial pad of the DUT to enter EngineerMode.
3. Select **DEBUG& LOG**, and then select **YLog**.
4. Select **Setting**, and then select **AUDIO SCENE**.
5. Click **Debug Setting & Tools**, **Modem To PC –> On**, as **Figure 1-17**.

NOTE

"*#*#83781#*#*" is the default secret code of EngineerMode on UNISOC platforms. If there is a need to modify, please contact related UNISOC engineers.

Figure 1-17 EngineerMode Setting



1.2.2 PC settings

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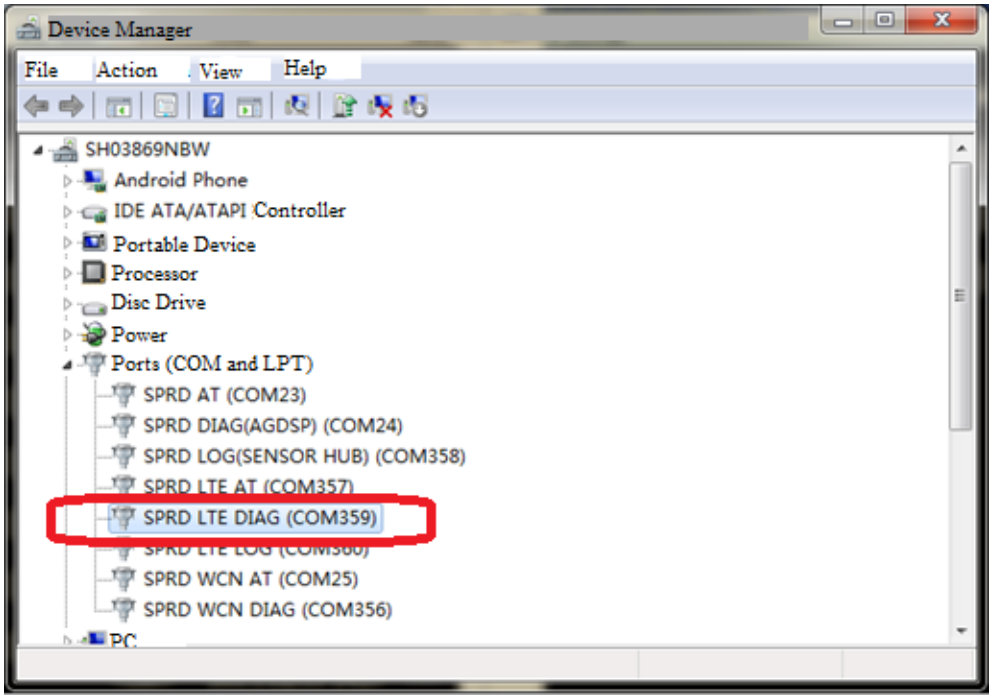
1. Use USB cable to connect DUT with PC.
2. Install UNISOC USB driver.
3. Open device manager to see whether there is DIAG ports. If the driver is properly installed, the COM port is shown in **Figure 1-18**.

Figure 1-18 DIAG ports



1.2.3 AudioTester settings

Configure as **1.1.2 Online Tuning**, and click **Connect**. If the connection is successful, Log window shows Connect With CS as **Figure 1-19**.

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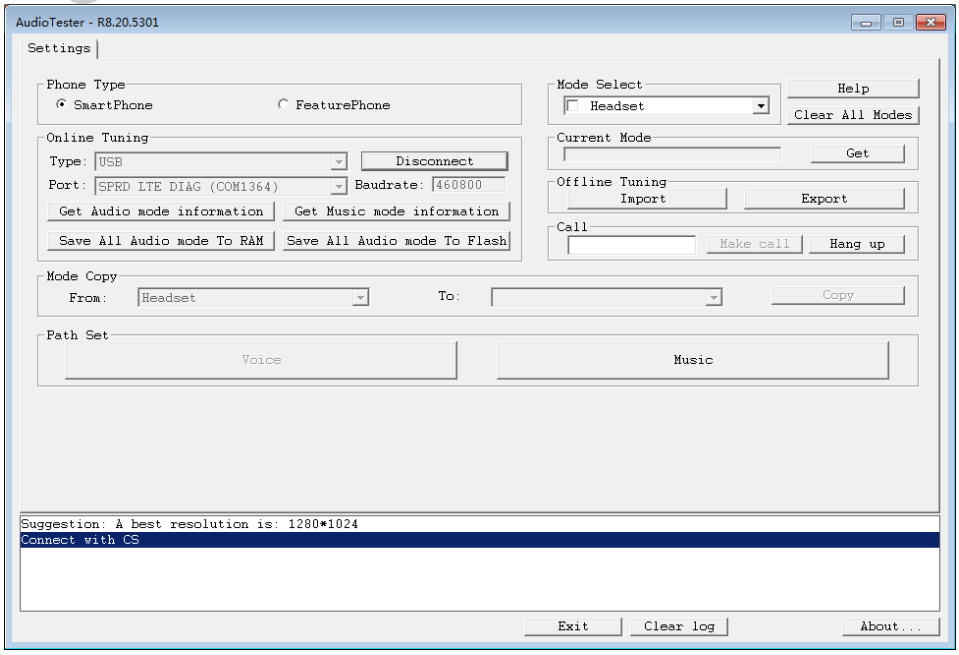
Figure 1-19 Connect successfully



- **Get Audio mode information**: enter Audio tuning interface.
- **Get Music mode information**: enter multimedia tuning interface.

1.3 Write&Read parameters

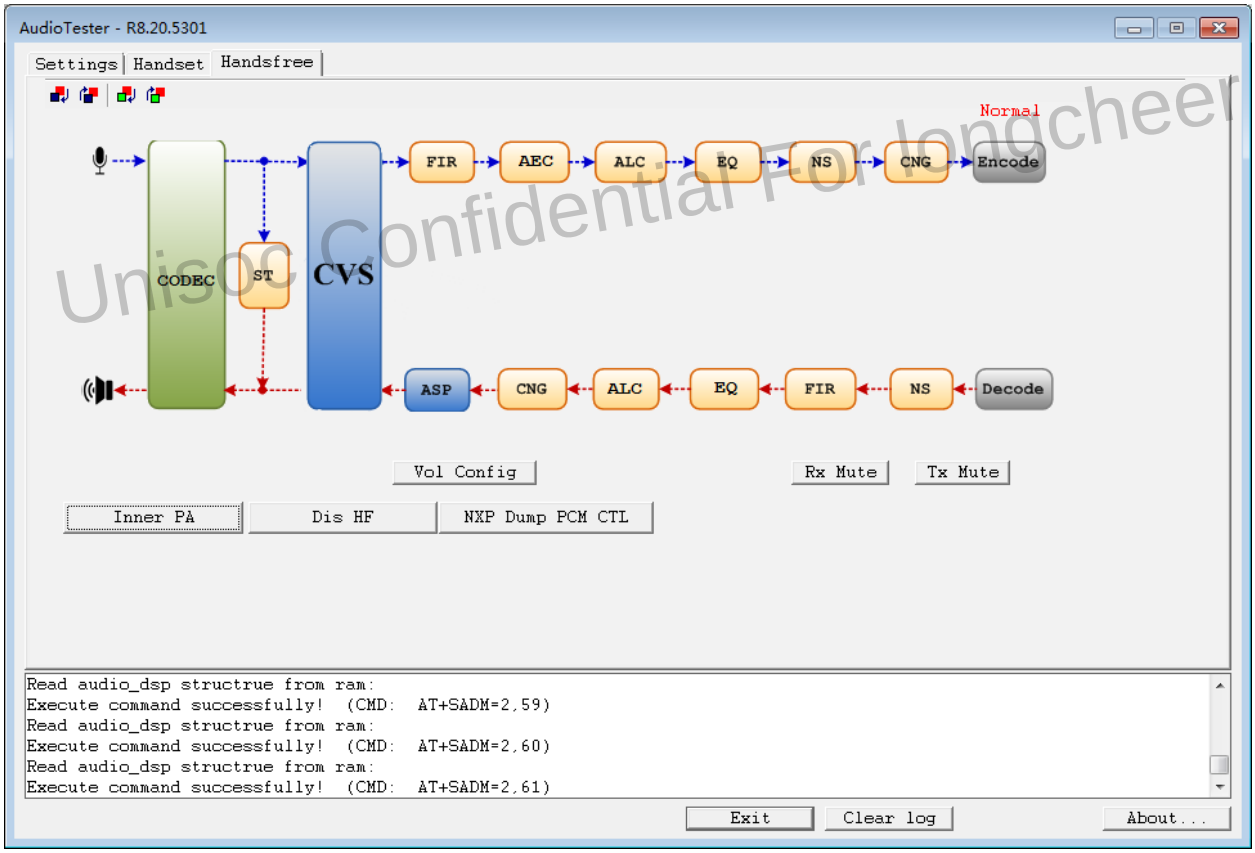
1.3.1 Audio Mode

In Audio Mode, there are four buttons to write parameters to DUT or read parameters from DUT.

Take handset mode as an example:

- : Write parameters to the RAM of DUT. The parameters will be lost after reboot.
- : Read parameters from the RAM of DUT.
- : Write parameters to the Flash of DUT. The parameters will not be lost after reboot.
- : Read parameters from the Flash of DUT.

Figure 1-20 Write & Read parameters in Audio Mode



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1.3.2 Music Mode

In Music mode, there are four buttons to write parameters to DUT or read parameters from DUT.

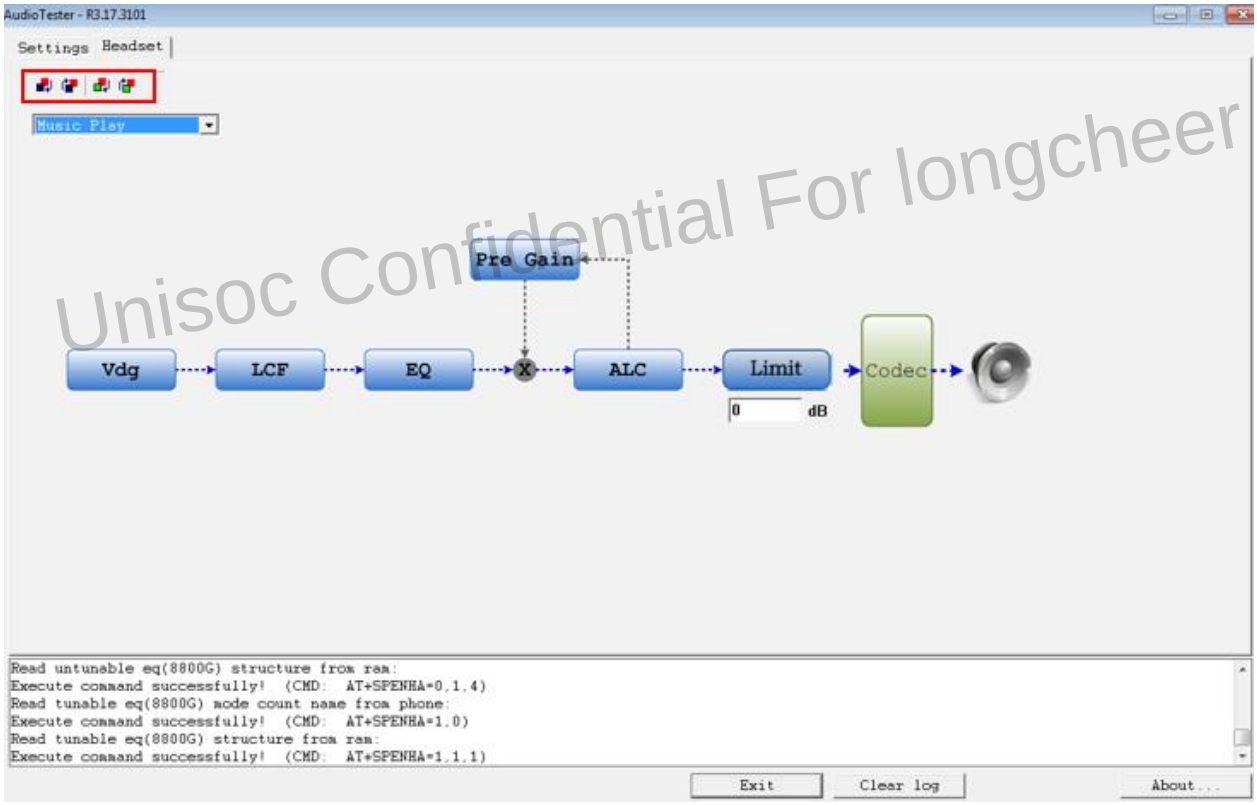
Feature phone

- : Write parameters to the RAM of DUT. Parameters will be effect immediately, but they will be lost after reboot.
- : Read parameters from the RAM of DUT.
- : Write parameters to the Flash of DUT. Parameters will not take effect immediately. They will not be lost after reboot and will take effect after reboot.
- : Read parameters from the Flash of DUT.

Smart phone

- & : When tuning music play, write parameters to RAM & Flash together.
- : Not use.
- : Read parameters from DUT.

Figure 1-21 Write & Read parameters in Music Mode



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NOTE

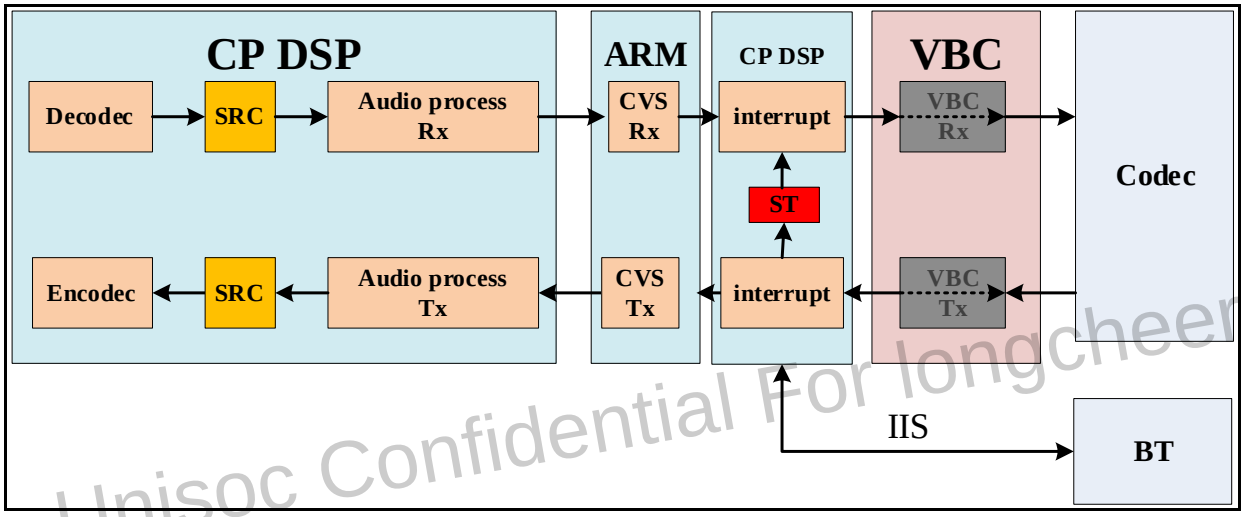
When tuning UNISOC's smartphone, after finishing FM tuning & voice record tuning, write parameters to RAM & Flash. Parameters will take effect after restarting FM or voice recording function.

2 Audio tuning

2.1 Voice Call

Voice Call includes call under different network modes, including scenarios as VOIP, Loopback, etc. Their data paths are the same as **Figure 2-1**.

Figure 2-1 Voice Call block



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NOTE

- Audio Process modules take effect in NB and WB modes.
- In SWB mode, Audio Process module doesn't take effect because of the insufficient DSP MIPS, and only CVS module takes effect.

2.1.1 CP DSP

DSP module is mainly responsible for the coding and decoding of vocoder, containing SRC and Audio Process module.

- Decodec/Encodec: vocoder coding and decoding, and AudioTester can't tune this module's parameter.
- SRC: sampling rate convertor between internet and Audio Process, take BTHSD8K(Bluetooth working in 8kHz sample rate) as an example:
  - BTHSD8K call in WCDMA_WB (16KHz) network mode, that is connecting the Bluetooth without echo noise reduction in WCDMA_WB network.
  - BT module =8KHz←→ Audio Process =8KHz ←→SRC←→ WCDMA_WB(16KHz)
- Audio Process: Audio effect algorithms in DSP call scenario.
- Interrupt: DSP interrupt. 20ms per frame.
- ST: SideTone module is in DSP interrupt, for details, see as **2.3.2 ST**.

2.1.2 ARM

Audio effect algorithms in ARM call scenario, mainly for echo and noise processing.

2.1.2.1 CVS

See **2.8 CVS**.

2.1.2.2 NXP

See **2.7 NXP**.

2.1.3 VBC

In Voice Call scenario, VBC is in Bypass status, and no parameters need to be tuned.

2.1.4 Codec

UNISOC matched PMIC, Audio Codec.

2.1.5 BT

Bluetooth module, need to configure IIS parameters, for details, see **5.12 IIS parameter configuration during calls**.

2.2 Voice Call interface

Take Voice Call Handsfree as an example, interface is shown in **Figure 2-2**.

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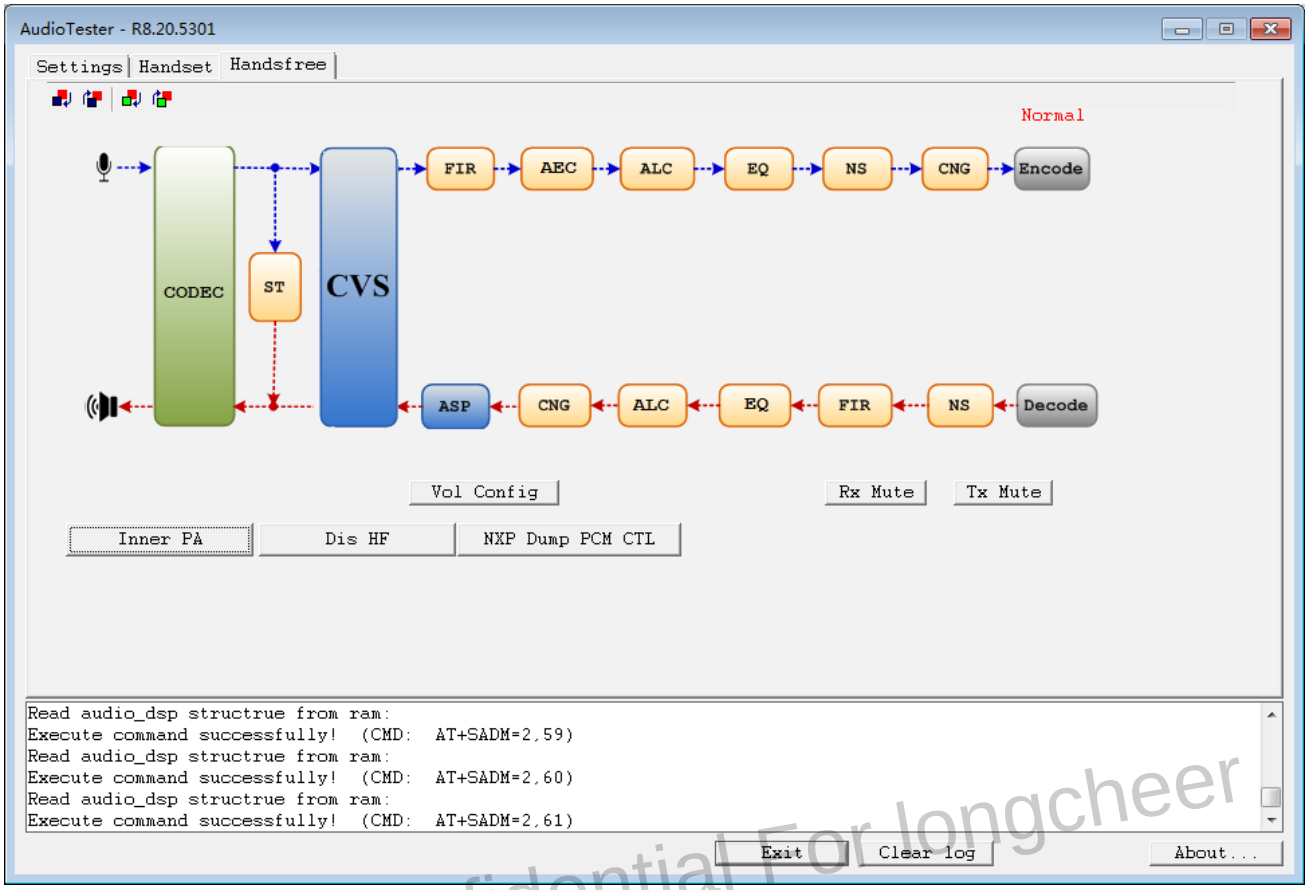
Figure 2-2 Voice Call Handsfree interface



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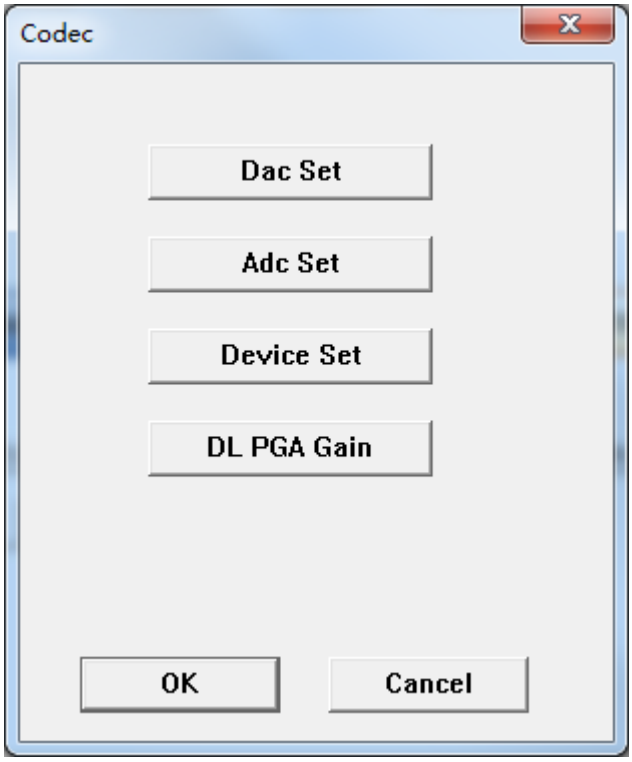
2.3 Voice Call Audio parameters

Voice Call\LoopBack\VOIP Audio processing modules are the same, and their Audio parameters results are the same. Take Voice Call Handsfree as an example.

2.3.1 Codec

Figure 2-3 Codec interface



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DAC set

DAC related configuration in Codec.

- Internal: use the platform codec shown in **Table 2-1**.

Table 2-1 DAC Set

Bit	Name	Description	Function	Note
bit4~0	DAC fs	Sample rate of DAC	• 0: 8K • 1: 12K • 2: 16K • Other: reserve	Software reservation. Not support.
bit5	DAC en	Enable of DAC	• 0: DAC off • 1: DAC on	NA
bit6	DACL en	Enable of DAC_L path	• 0: DAC_L path off • 1: DAC_L path on	NA
bit7	DACR en	Enable of DAC_R path	• 0: DAC_R path off • 1: DAC_R path on	NA

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- External: use the third-party codec, not used currently.

ADC set

ADC related configuration in Codec.

- Internal: use the platform codec shown in **Table 2-2**.

Table 2-2 ADC set

Bit	Name	Description	Function	Remark
bit4~0	ADC fs	Sample rate of ADC	• 0: 8K • 1: 12K • 2: 16K • Other: reserve	Software reservation. Not support.
bit5	ADC en	Enable of ADC	• 0: ADC off • 1: ADC on	NA
bit6	ADCL en	Enable of ADC_L path	• 0: ADC_L path off • 1: ADC_L path on	NA
bit7	ADCR en	Enable of ADC_R path	• 0: ADC_R path off • 1: ADC_R path on	NA

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- External: use the third-party codec, not used currently.

Device Set (Path Set)

Path set in Codec.

- Internal: use the platform codec shown in **Table 2-3**.

Table 2-3 Path Set

Bit	Name	Description	Function
bit0	HP	HeadPhone switch	• 0: off • 1: on
bit1	HP_L	HeadPhone L path switch	• 0: off • 1: on
bit2	HP_R	HeadPhone R path switch	• 0: off • 1: on
bit3	SPK	Speaker switch	• 0: off • 1: on
bit4	SPK_L	Speaker L path switch	• 0: off • 1: on

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Bit	Name	Description	Function
bit5	SPK_R	Speaker R path switch	• 0: off • 1: on
bit6	RCV	Receiver switch	• 0: off • 1: on
bit7	RCV_L	Receiver L path switch	• 0: off • 1: on
bit8	RCV_R	Receiver R path switch	• 0: off • 1: on
bit9	MIC	MIC switch	• 0: off • 1: on
bit10	MIC_L	MIC L path switch	• 0: off • 1: on
bit11	MIC_R	MIC R path switch	• 0: off • 1: on
bit12	HPMIC	earphone mic path switch	• 0: off • 1: on
bit13	Line_in	Line_in switch	• 0: off • 1: on
bit14	Line_in_L	Line_in L path switch	• 0: off • 1: on
bit15	Line_in_R	Line_in R path switch	• 0: off • 1: on

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- External: use the third-party codec, not used currently.

DL PGA Gain

Not take effect, code unsupported.

2.3.2 ST

Figure 2-4 Side Tone

ST(Side Tone)

- Off: Close Side Tone
- DSP Digital: Open Side Tone path

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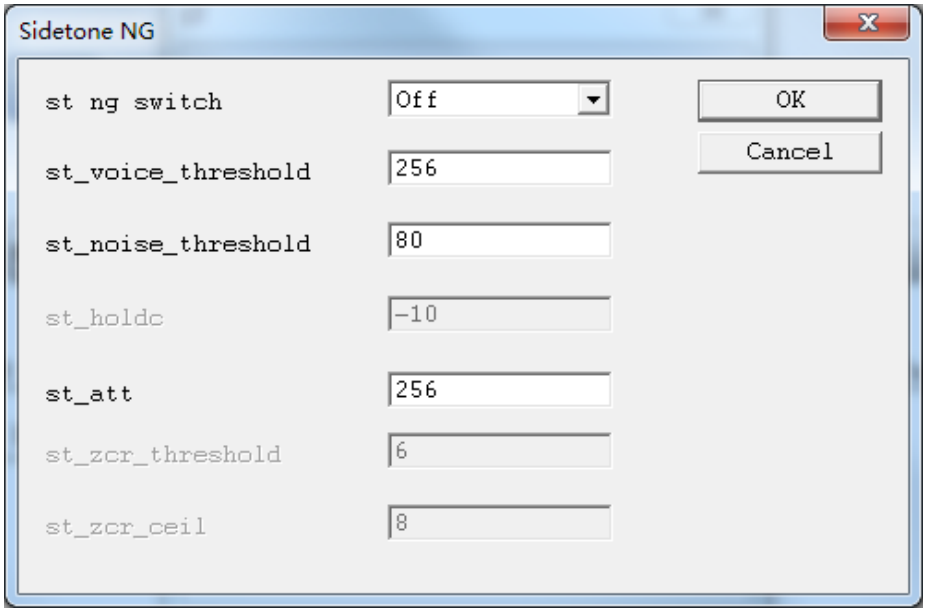
Side Tone NG

Figure 2-5 Side Tone NG



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- st ng switch: NG switch On/Off
- st_voice_threshold: If the signal level larger than st_voice_threshold, NG doesn't take effect. Value range: 0~32767.
- st_noise_threshold: If the signal level smaller than st_noise_threshold, NG doesn't take effect. Value range: 0~32767.
- st_att: NG function of attenuating noise. Range:0~4096, 20*log(4096/st_att) dB
  - 4096->0dB
  - 2048->6dB
  - 256->24dB

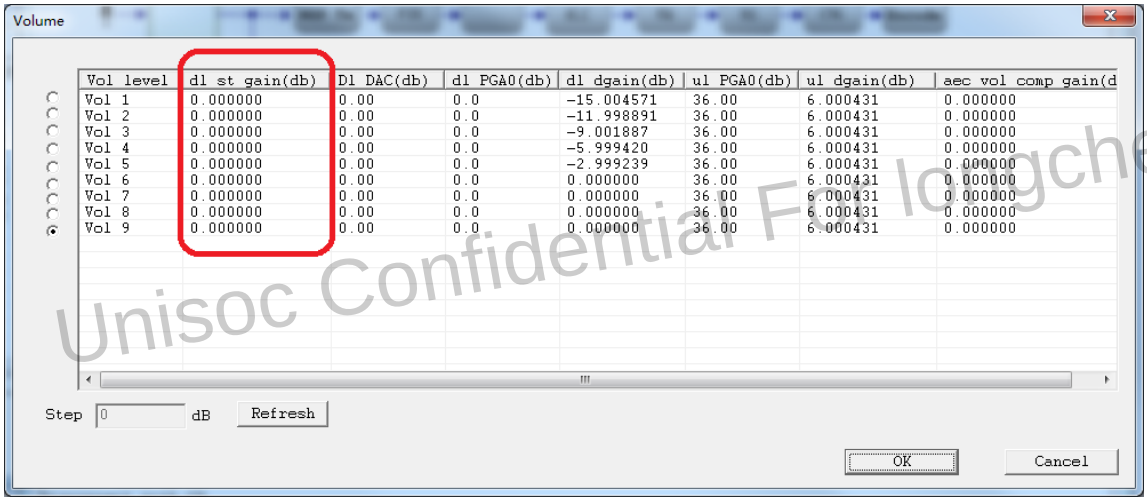
2.3.2.1 ST FIR

Not take effect.

2.3.2.2 Side Tone Gain

For details, see dl_st_gain in **2.3.10 Vol Config**.

Figure 2-6 dl_st_gain



Vol level	dl st gain(db)	Dl DAC(db)	dl PGA0(db)	dl dgain(db)	ul PGA0(db)	ul dgain(db)	aec vol comp gain(d
Vol 1	0.000000	0.00	0.0	-15.004571	36.00	6.000431	0.000000
Vol 2	0.000000	0.00	0.0	-11.998891	36.00	6.000431	0.000000
Vol 3	0.000000	0.00	0.0	-9.001887	36.00	6.000431	0.000000
Vol 4	0.000000	0.00	0.0	-5.999420	36.00	6.000431	0.000000
Vol 5	0.000000	0.00	0.0	-2.999239	36.00	6.000431	0.000000
Vol 6	0.000000	0.00	0.0	0.000000	36.00	6.000431	0.000000
Vol 7	0.000000	0.00	0.0	0.000000	36.00	6.000431	0.000000
Vol 8	0.000000	0.00	0.0	0.000000	36.00	6.000431	0.000000
Vol 9	0.000000	0.00	0.0	0.000000	36.00	6.000431	0.000000

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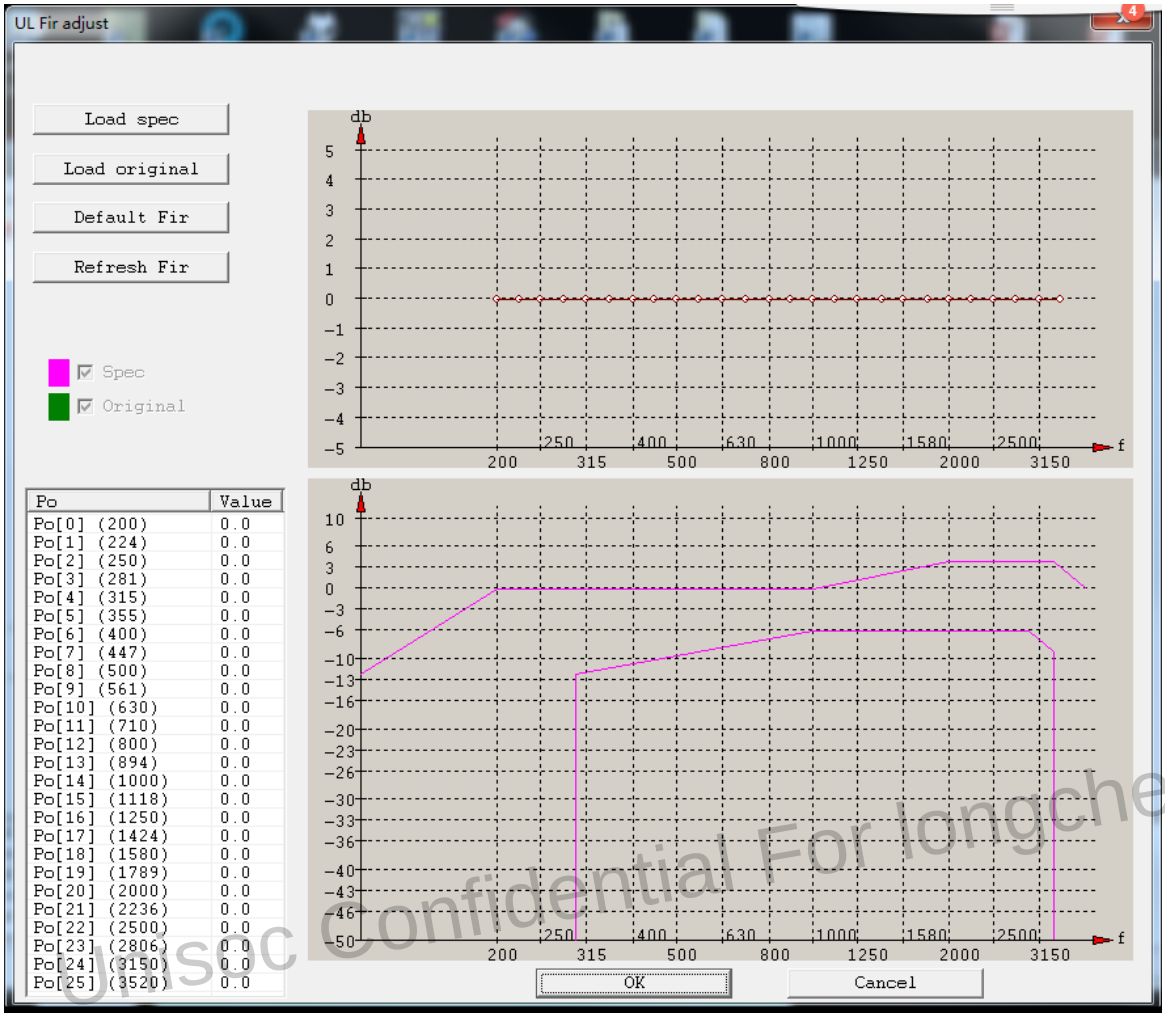
2.3.3 FIR

FIR: Finite impulse response.

FIR exists in both Tx and Rx path with the same parameter definitions. Bandwidth in NB mode is 0~4KHz, WB mode is 0~8KHz.

Figure 2-7 FIR



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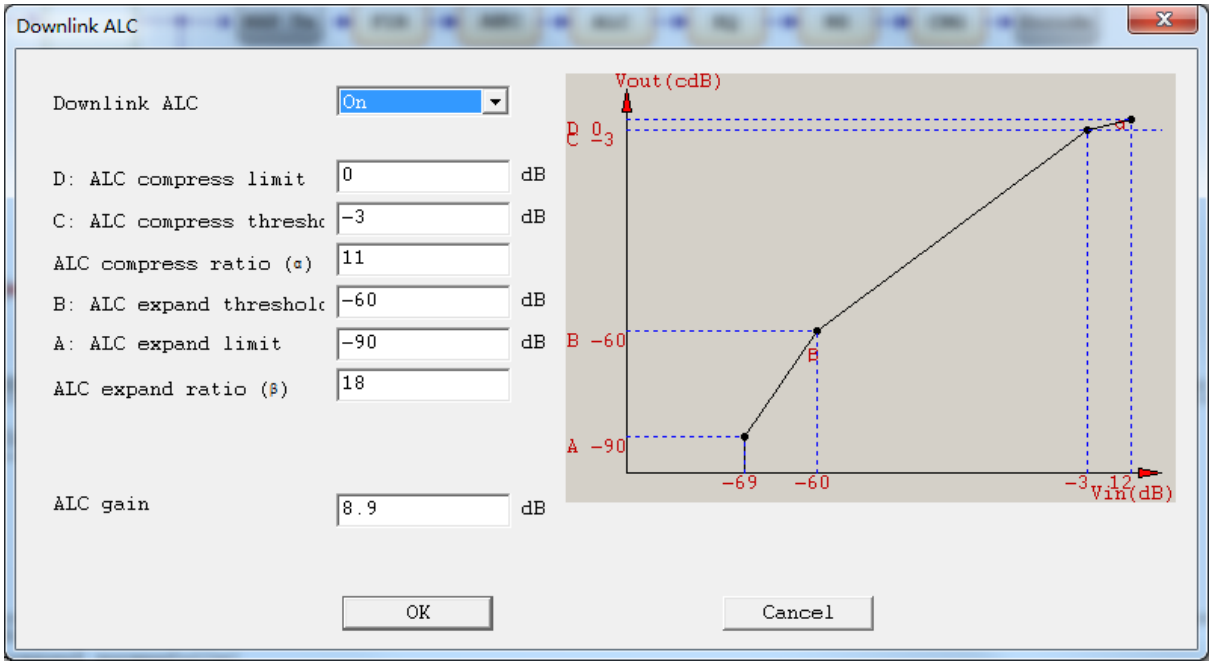
- Default Fir: Bypass FIR filter.
- Refresh Fir: click it to get the updated curve after dragging the gain of each frequency.

2.3.4 ALC

Figure 2-8 ALC



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- ALC: ALC switch On/Off
- ALC compress limit: when the input signal level is larger than ALC compress threshold, the signal will be compressed.
- ALC compress threshold: when the input signal level is larger than ALC compress threshold, the output signal will be compressed. If the signal is larger than ALC compress limit, the output signal will be limited to ALC_ compress _ limit.
- ALC compress ratio: ΔOutput gain/ΔInput gain
- ALC expand threshold: when the input signal level is lower than the ALC_ expand _ threshold, the signal will be expanded.
- ALC expand limit: when the input signal level is lower than the ALC_ expand _ threshold, the output signal will be expanded. If the signal is lower than ALC_ expand _ limit, the output signal will be limited to ALC_ expand _ limit.
- ALC expand ratio: ΔOutput gain/ΔInput gain
- ALC gain: The pre-gain before ALC, and the gain will be controlled by ALC.

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2.3.5 EQ(IIR Filter)

Figure 2-9 Tx EQ IIR Filter

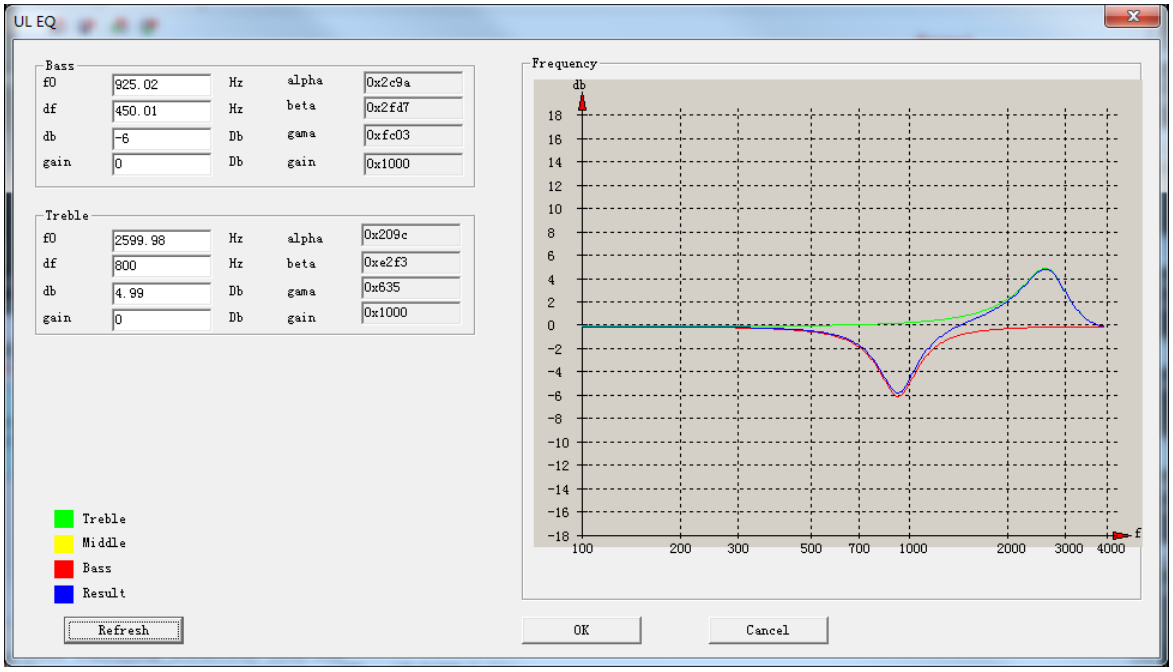

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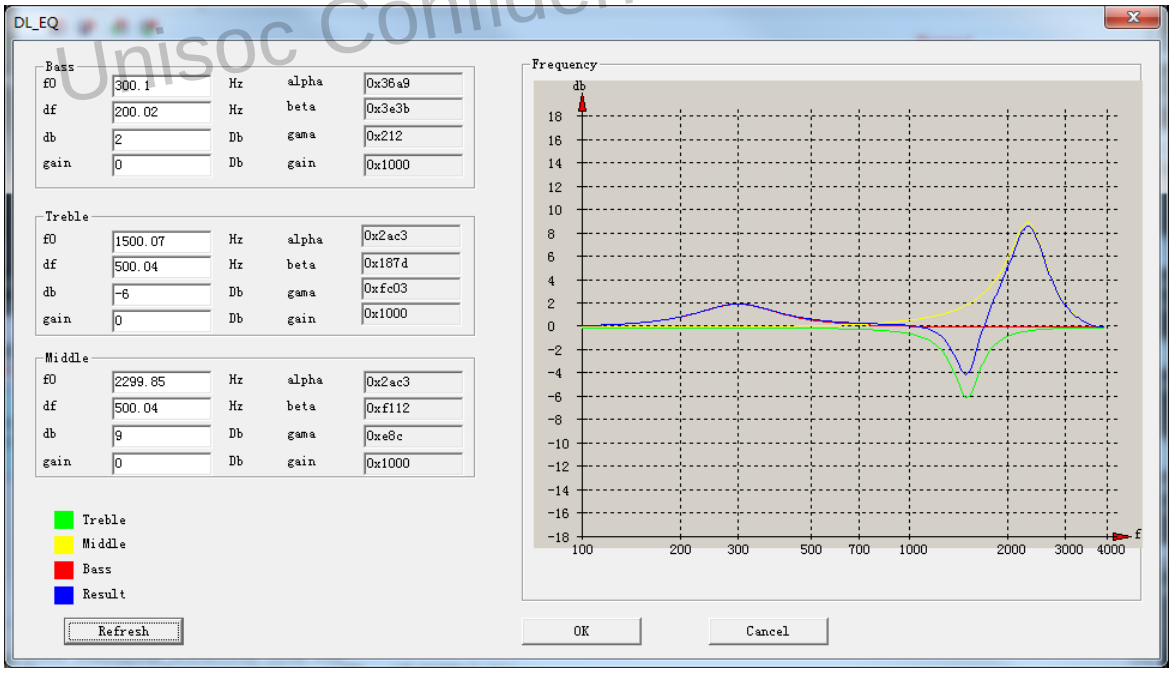
Figure 2-10 Rx EQ IIR Filter



- f0: Center Frequency
- df: bandwidth (the peak value drop 3dB)

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- db: boost in dB at central frequency
- Gain: basic gain in dB (all frequency gain)

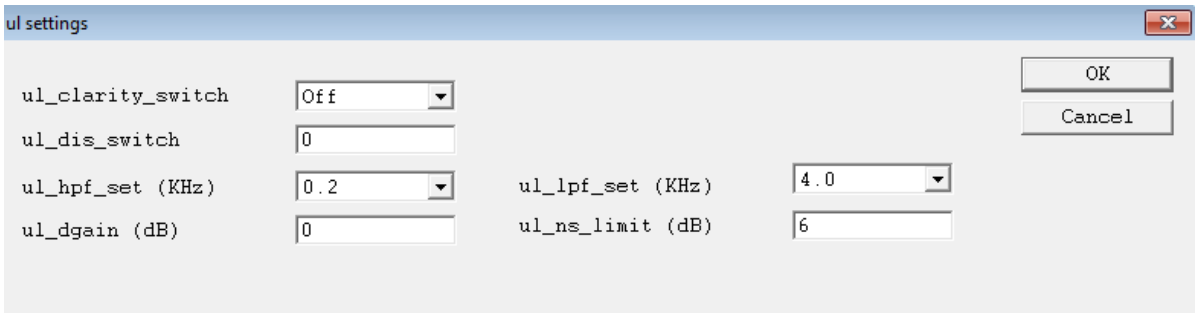
NOTE

When tuning SFR, tuning Tx EQ (IIR) parameters first, and then Tx FIR parameters.

When tuning RFR, tuning Rx EQ (IIR) parameters first, and then Rx FIR parameters.

2.3.6 NS(Noise Suppression or Voice Clarity)

Figure 2-11 ul settings



- Clarity switch: 1 = on; 0 = off
- dis_switch: distortion suppression switch
- hpf_set: high pass filter cut-off frequency
- lpf_set: low pass filter cut-off frequency
- dgain: digital gain
- ns_limit: noise suppression

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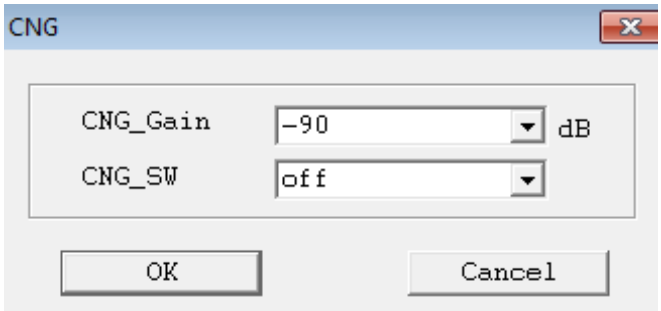
2.3.7 CNG

Figure 2-12 CNG



- CNG_Gain: The level of comfortable noise generated by CNG function.
- CNG_SW: CNG function switch.

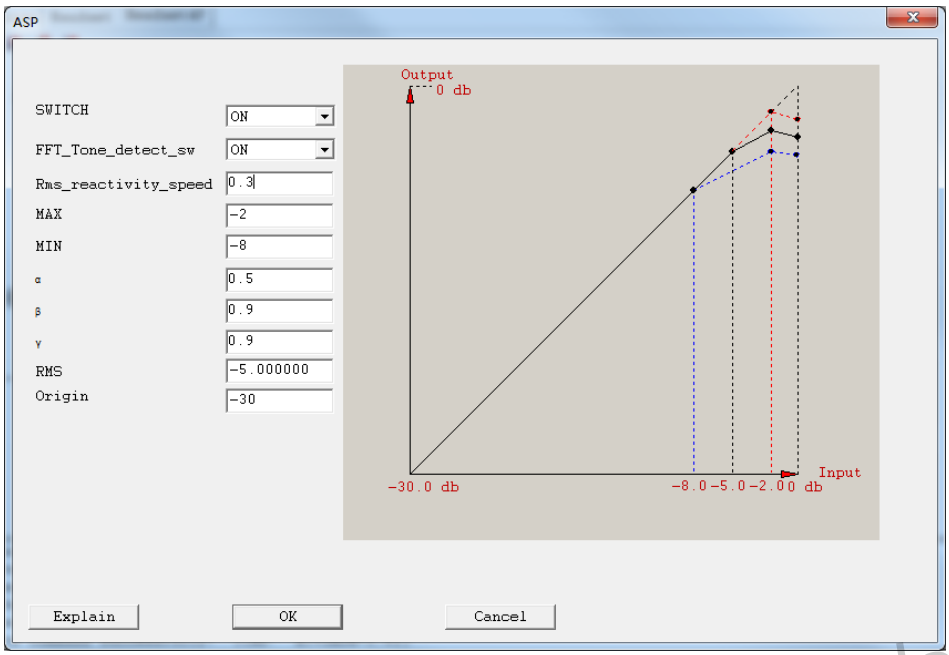
2.3.8 ASP

ASP (Acoustic Shock Protection) is only used in NXP Algorithm platform, and CVS can't use.

Tone protection function: when the other sends Tones such as TDMF in Call scenario, the volume of Tone follows the volume of voice.

ASP default parameters are shown in **Figure 2-13**.

Figure 2-13 ASP default parameters



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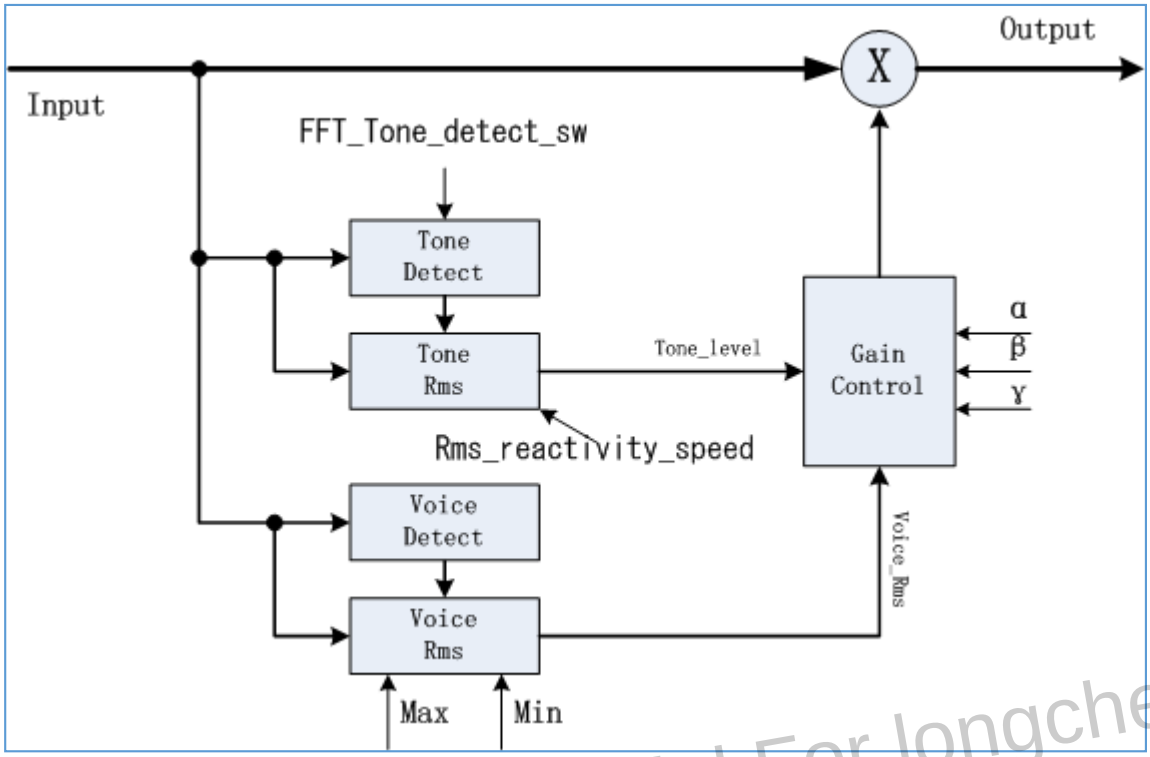
2.3.8.1 ASP Block

Figure 2-14 ASP block



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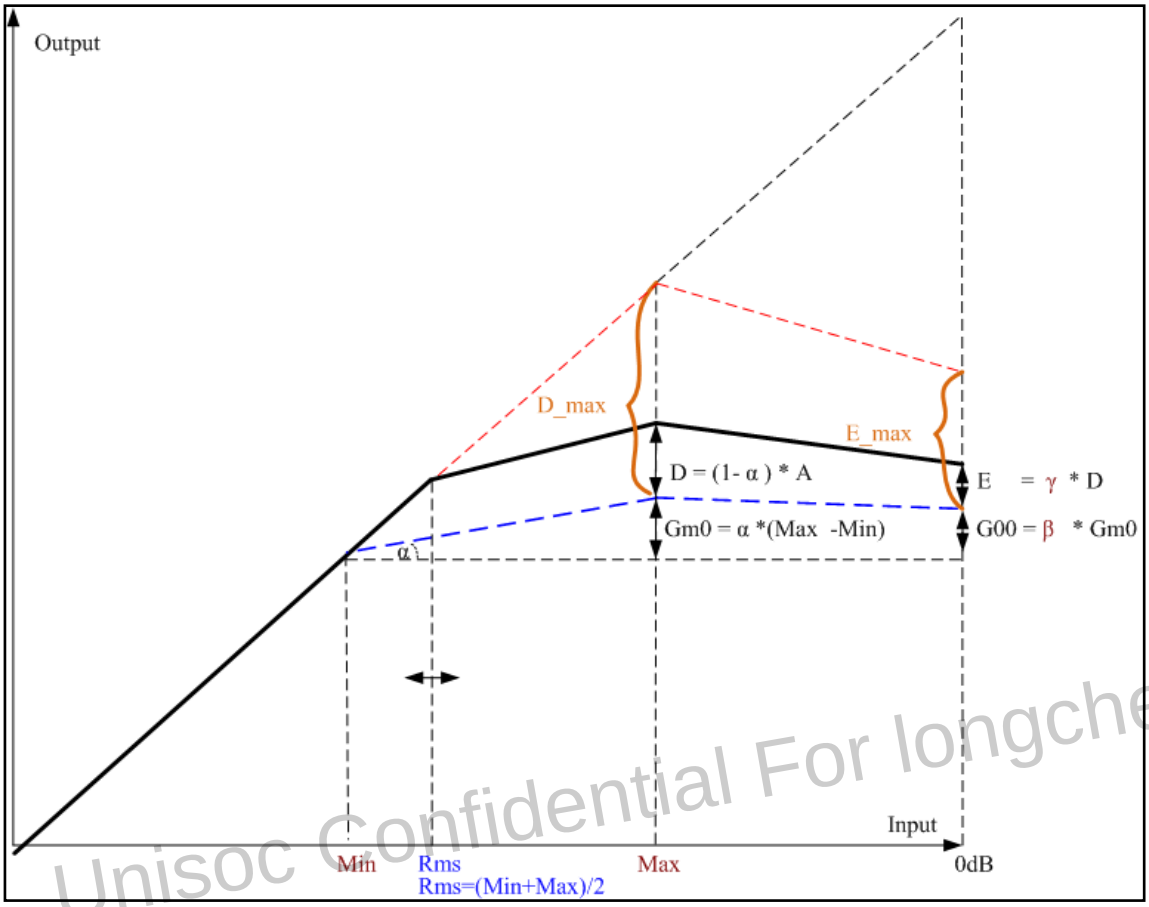
- Tone Detect: Check the input signal is Tone or Voice. It will detect detail when FFT_Tone_detect_sw set on.
- Tone Rms: Detect the value of Tone. Rms_reactivity_speed controls Rms updata rate.
- Voice Detect: Check the input signal is Tone or Voice.
- Voice Rms: Detect the value of Voice.
- Gain Control: Control Tone value.

2.3.8.2 ASP Principle

Figure 2-15 ASP principle



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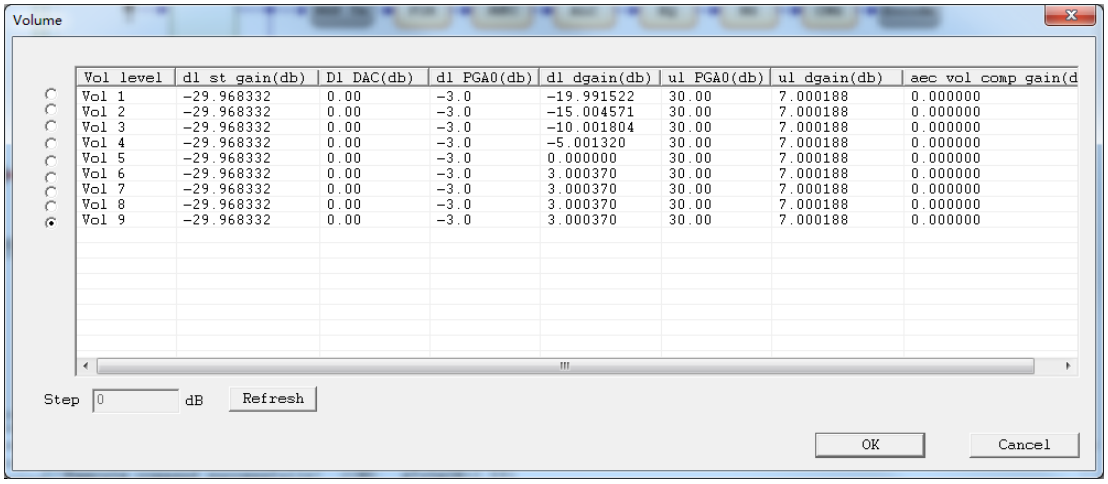
- Draw the line according to Max, Min, α, β, γ, Rms.
- Rms (default value) = (Max+Min)/2

2.3.9 NXP_VB_CTL

The default parameter is 0x1111.

2.3.10 Vol Config

Figure 2-16 Volume



Vol level	dl st gain(db)	Dl DAC(db)	dl PGA0(db)	dl dgain(db)	ul PGA0(db)	ul dgain(db)	aec vol comp gain(d
Vol 1	-29.968332	0.00	-3.0	-19.991522	30.00	7.000188	0.000000
Vol 2	-29.968332	0.00	-3.0	-15.004571	30.00	7.000188	0.000000
Vol 3	-29.968332	0.00	-3.0	-10.001804	30.00	7.000188	0.000000
Vol 4	-29.968332	0.00	-3.0	-5.001320	30.00	7.000188	0.000000
Vol 5	-29.968332	0.00	-3.0	0.000000	30.00	7.000188	0.000000
Vol 6	-29.968332	0.00	-3.0	3.000370	30.00	7.000188	0.000000
Vol 7	-29.968332	0.00	-3.0	3.000370	30.00	7.000188	0.000000
Vol 8	-29.968332	0.00	-3.0	3.000370	30.00	7.000188	0.000000
Vol 9	-29.968332	0.00	-3.0	3.000370	30.00	7.000188	0.000000

- Vol Level: volume level
- dl_st_gain: Side Tone digital gain
- DL_DAC: Codec DAC analog gain, suggest to set the default gain 0dB
- Dl_PGA: Codec downlink path analog gain
- Dl_dgain: downlink path digital gain

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Smart Phone

Table 2-4 Description of Smart Phone volume

Volume	NV
1	AUDIO\AUDIO_DSP\XXX\AudioStructure\extend\extend[110]
…	…
5	AUDIO\AUDIO_DSP\XXX\AudioStructure\extend\extend[114]
6	AUDIO\AUDIO_DSP\XXX\AudioStructure\extend\extend[115]
7	AUDIO\AUDIO_DSP\XXX\AudioStructure\extend\extend[116]
8	AUDIO\AUDIO_DSP\XXX\AudioStructure\extend\extend[117]
9	AUDIO\AUDIO_DSP\XXX\AudioStructure\extend\extend[118]

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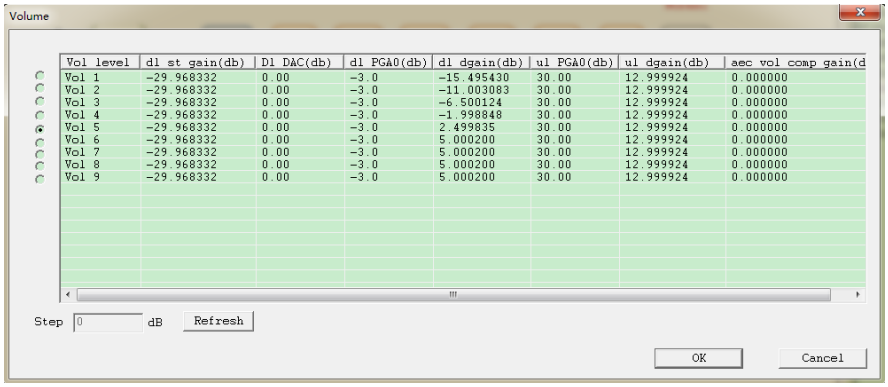
Table 2-5 Smart phone volume



Volume

Vol level	dl st gain(db)	Dl DAC(db)	dl PGA0(db)	dl dgain(db)	ul PGA0(db)	ul dgain(db)	aec vol comp gain(d
Vol 1	-29.968332	0.00	-3.0	-15.495430	30.00	12.999924	0.000000
Vol 2	-29.968332	0.00	-3.0	-11.003083	30.00	12.999924	0.000000
Vol 3	-29.968332	0.00	-3.0	-6.500124	30.00	12.999924	0.000000
Vol 4	-29.968332	0.00	-3.0	-1.998848	30.00	12.999924	0.000000
Vol 5	-29.968332	0.00	-3.0	2.499835	30.00	12.999924	0.000000
Vol 6	-29.968332	0.00	-3.0	5.000200	30.00	12.999924	0.000000
Vol 7	-29.968332	0.00	-3.0	5.000200	30.00	12.999924	0.000000
Vol 8	-29.968332	0.00	-3.0	5.000200	30.00	12.999924	0.000000
Vol 9	-29.968332	0.00	-3.0	5.000200	30.00	12.999924	0.000000

Step 0 dB Refresh

OK Cancel

Feature phone

Table 2-6 Description of Feature Phone volume

Volume	NV
1	None(Base on level 2, step decrease)
…	…
6	None(Base on level 7, step decrease)
7	audio\audio\XXX\AudioStructure\extend\extend[110]
8	audio\audio\XXX\AudioStructure\extend\extend[111]
…	…
15	audio\audio\XXX\AudioStructure\extend\extend[118]

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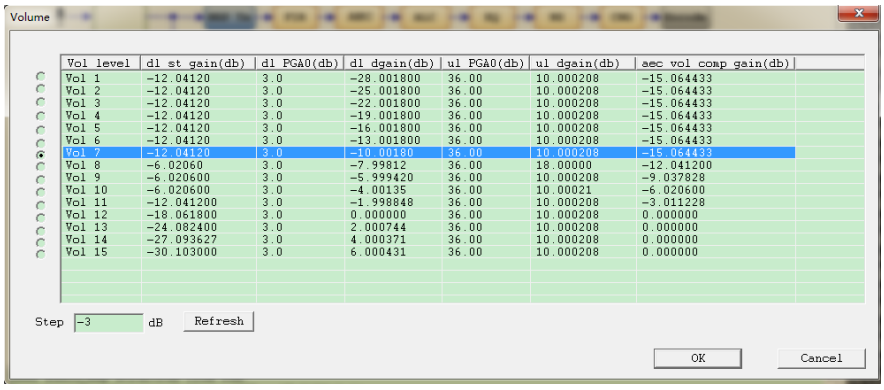
Figure 2-17 Feature Phone volume



Volume

Vol level	dl st gain(db)	dl PGA0(db)	dl dgain(db)	ul PGA0(db)	ul dgain(db)	aec vol comp gain(db)
Vol 1	-12.04120	3.0	-28.001800	36.00	10.000208	-15.064433
Vol 2	-12.04120	3.0	-25.001800	36.00	10.000208	-15.064433
Vol 3	-12.04120	3.0	-22.001800	36.00	10.000208	-15.064433
Vol 4	-12.04120	3.0	-19.001800	36.00	10.000208	-15.064433
Vol 5	-12.04120	3.0	-16.001800	36.00	10.000208	-15.064433
Vol 6	-12.04120	3.0	-13.001800	36.00	10.000208	-15.064433
Vol 7	-12.04120	3.0	-10.00180	36.00	10.000208	-15.064433
Vol 8	-6.02060	3.0	-7.99812	36.00	18.00000	-12.041200
Vol 9	-6.020600	3.0	-5.999420	36.00	10.000208	-9.037828
Vol 10	-6.020600	3.0	-4.00135	36.00	10.00021	-6.020600
Vol 11	-12.041200	3.0	-1.998848	36.00	10.000208	-3.011228
Vol 12	-18.061800	3.0	0.000000	36.00	10.000208	0.000000
Vol 13	-24.082400	3.0	2.000744	36.00	10.000208	0.000000
Vol 14	-27.093627	3.0	4.000371	36.00	10.000208	0.000000
Vol 15	-30.103000	3.0	6.000431	36.00	10.000208	0.000000

Step -3 dB Refresh

OK Cancel

- Ul_PGA: Codec uplink path analog gain
- Ul_dgain: uplink path digital gain

- Aec_vol_comp_gain: not take effect

2.3.11 Rx Mute

For debug, downlink path mute.

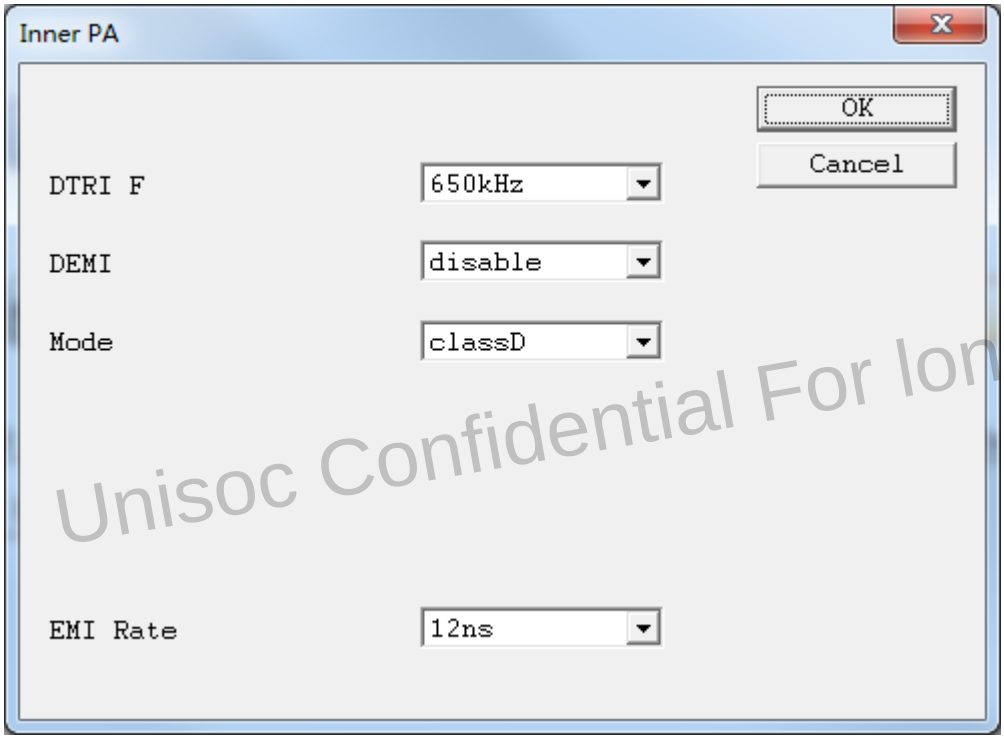
2.3.12 Tx Mute

For debug, uplink path mute.

2.3.13 Inner PA

SPK PA parameters configuration

Figure 2-18 Inner PA



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Class AB/D configuration:

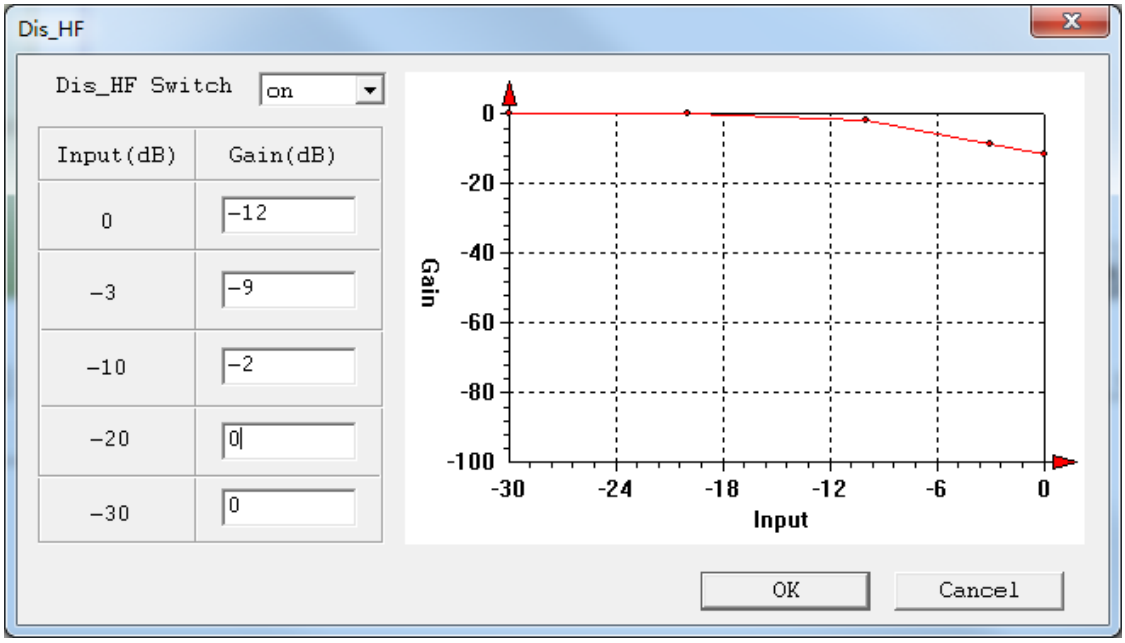
- DTRI_F: switch frequency of classD
- DEMI: EMI switch of classD
- Mode: select class AB/D
- EMI Rate: select EMMI EMI rate

As EMI radiation points and energies of classD vary with different configurations, tuning when FM/GPS/ RF receiving interference.

2.3.14 Dis HF

Distortion optimization module

Figure 2-19 Dis HF



- Dis_HF_Switch: switch of distortion optimization module.
- Input (dB): the amplitude of input signal.
- Gain (dB): the gain of corresponding input amplitude.

2.3.15 NXP Dump PCM CTL

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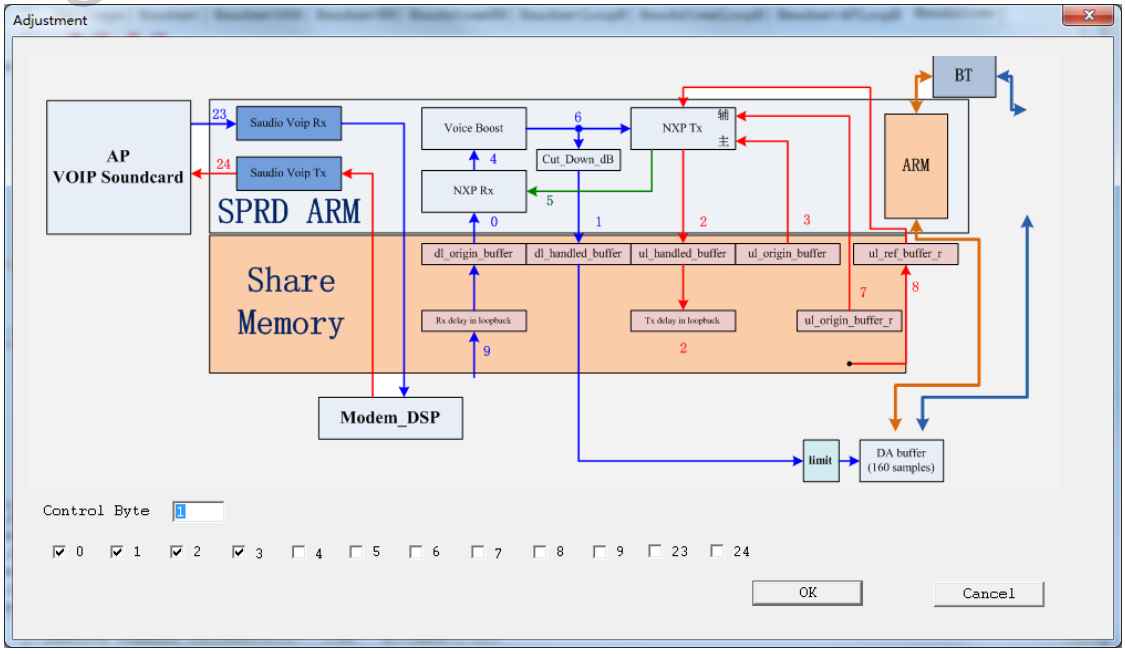
Figure 2-20 Audio PCM Dump



Control Byte

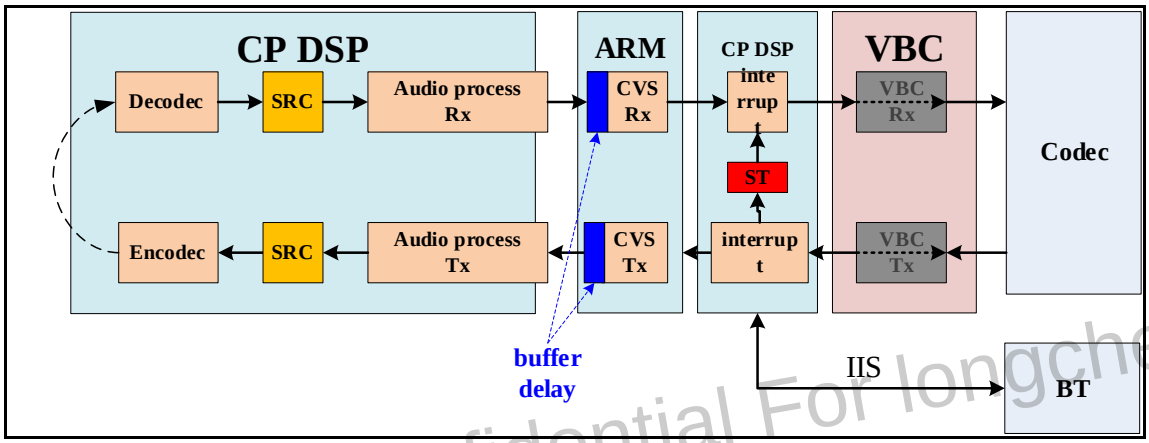
- 1: Dump function is on
- 0: Dump function is off

Refer to the data in the graphical display of the tool for each dump point.

2.4 Loopback

The audio effect processing of Loopback is basically the same as that of Voice Call. Buffer Delay is used to control the delay time, which can be set in the tool interface.

Figure 2-21 Loopback block



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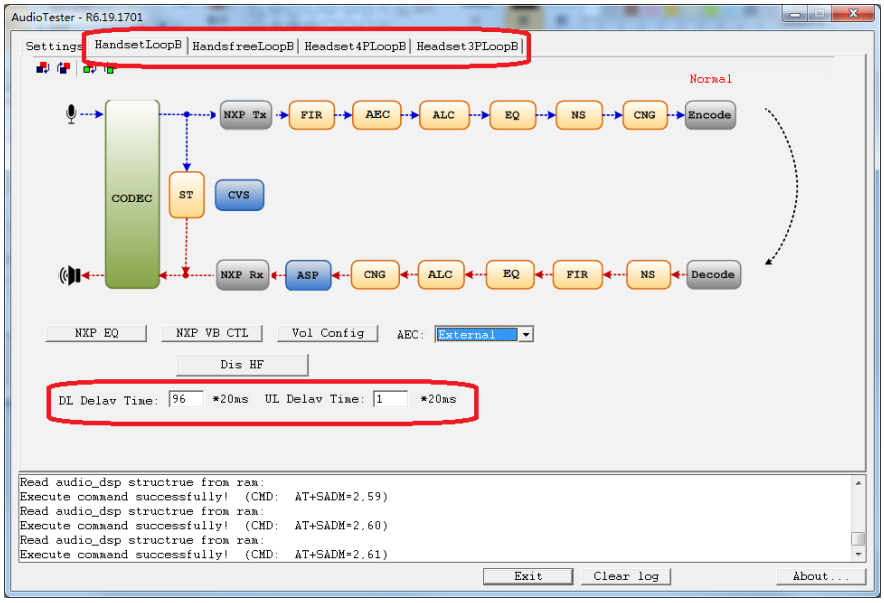
Loopback tuning interface is shown in **Figure 2-22**.

Figure 2-22 Loopback tuning interface



Loopback supports the following four modes:

- HandsetLoopB: Loopback in handset mode
- HandsfreeLoopB: Loopback in handsfree mode
- Headset4PLoopB: Loopback in 4 pole earphone mode
- Headset3PLoopB: Loopback in 3 pole earphone mode

NOTE

Only HandsetLoopB parameter can set Loopback Delay (Buffer Delay), which is suitable for all LoopBack scenes after modification.

NV parameters address: The parameters in NV, audio_arm\audio_arm\Handset\AudioStructure\reserve\reserve[63]

bit0~7: downlink buffer delay

bit8~15: uplink buffer delay

default value is 0x160=0b 0000，0001，0110，0000

Downlink buffer delay = 96

Uplink buffer delay = 1

2.5 VOIP

VOIP audio process is the same as Voice Call.

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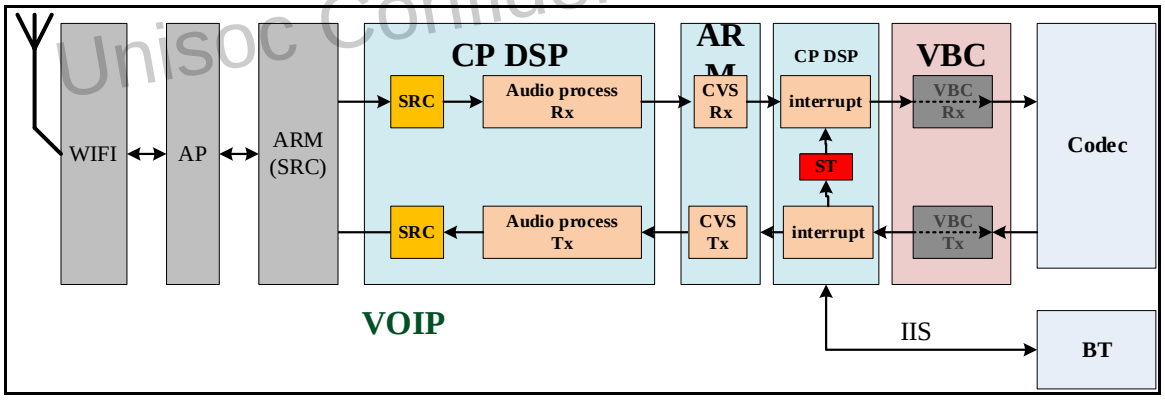
Figure 2-23 VOIP block



The data flow of VOIP is the same as that of Voice Call/. For parameter tuning, see **2.3 Voice Call Audio parameters**. VOIP tuning interface is shown in **Figure 2-24**.

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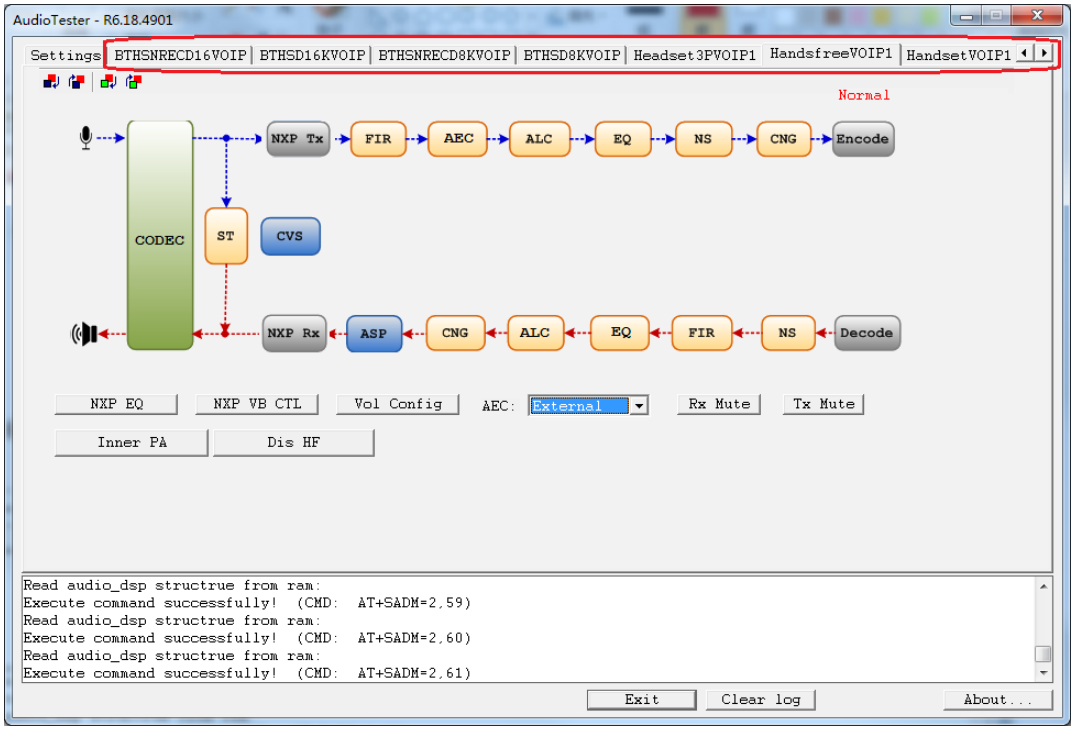
Figure 2-24 VOIP tuning interface



For VOIP parameters description, see **Table 2-7**.

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Table 2-7 VOIP parameters description

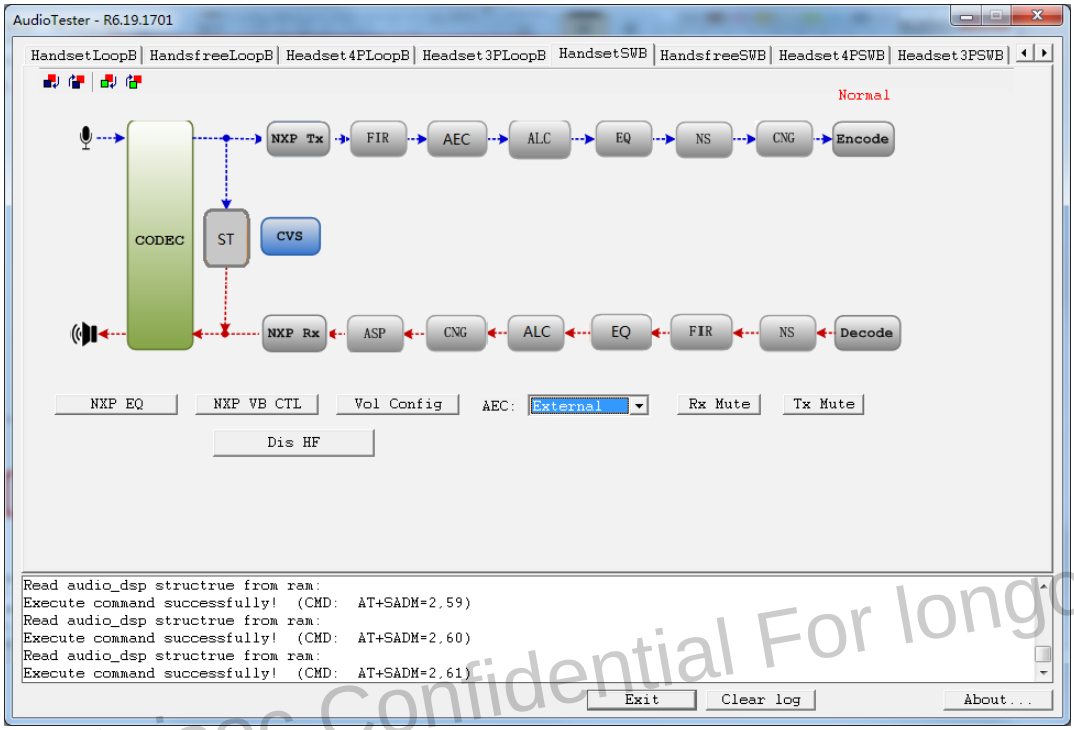
Parameter	Description	Sample rate
HandsetVOIP1	receiver mode for VOIP Call	Audio effect processing 16KHz sample rate
HandsfreeVOIP1	speaker mode for VOIP Call	Audio effect processing 16KHz sample rate
Headset4PVOIP1	4 pole earphone mode for VOIP Call	Audio effect processing 16KHz sample rate
Headset3PVOIP1	3 pole earphone mode for VOIP Call	Audio effect processing 16KHz sample rate
BTHSD8KVOIP1	8K BTHS mode for VOIP Call	Audio effect processing 8KHz sample rate
BTHSNRECD8KVOIP1	8K BTHSNREC mode for VOIP Call	Audio effect processing 8KHz sample rate
BTHSD16KVOIP1	16K BTHS mode for VOIP Call	Audio effect processing 16KHz sample rate
BTHSNRECD16KVOIP1	16K BTHSNREC mode for VOIP Call	Audio effect processing 16KHz sample rate

For audio effect processing in VOIP scene, DSP converts the sampling rate to 16KHz, except for 8KHz sampling in Bluetooth mode.

2.6 VoLTE SWB

For VoLTE SWB (Super Wide Band), DSP Audio Process doesn't take effect in 32KHz sample rate, and only uplink and downlink digital gain in Vol Config are saved. Audio parameters can refer to CVS module, see **2.8 CVS**.

Figure 2-25 SWB tuning interface



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2.7 NXP

Currently, NXP is only used in platforms as SC6531, SC7701, SC7703, etc. Others use CVS.

Refer to **5.3 NXP tuning** for a simple tuning, and the detailed tuning please contact UNISOC engineer support.

2.8 CVS

CVS (Clear Voice Suite) is UNISOC designed echo noise clear algorithm.

Supporting:

- Handset: mono/dual MIC
- Handsfree: mono MIC

For parameters tuning, please refer to the corresponding chip platform Audio Tuning Guide in Uni-support of UNISOC. If you have any questions, please contact UNISOC engineers.

3 Multimedia tuning

3.1 Music Play tuning interface

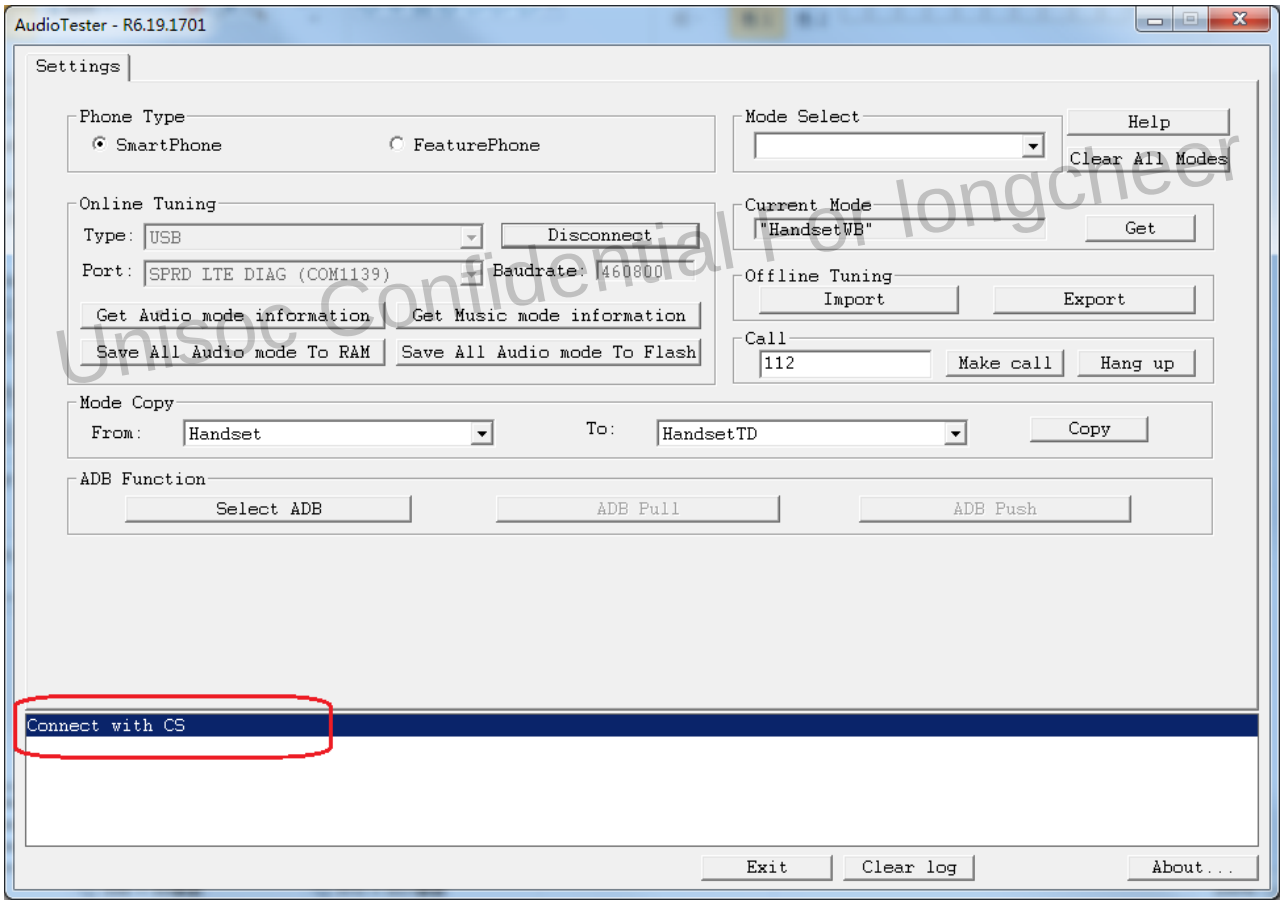
The connection process for reading DUT music mode parameters in the music play tuning interface is as follows.

Step 1 Select **SmartPhone** or **FeaturePhone** on the settings interface.

Step 2 Select **SPRD LTE DIAG** port, and click **Connect** button.

If DUT and the tool are successfully connected, the log window at the bottom of the tool interfaces **Connect with CS**, as shown in **Figure 3-1**.

Figure 3-1 DUT connection



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Step 3 Click **Get Music mode information** button to enter the multimedia debugging interface.

Reading parameters such as Music Play, FM, Record (SmartPhone), and AP/ARM. FeaturePhone Record channel is the same as Call.

----End

3.2 Music Mode

You can read parameters of RAM and Flash in Music Mode interface as shown in **Figure 3-2**. Icons and corresponding function description is shown in **Table 3-1**.

Figure 3-2 Read parameters of RAM and Flash in Music Mode

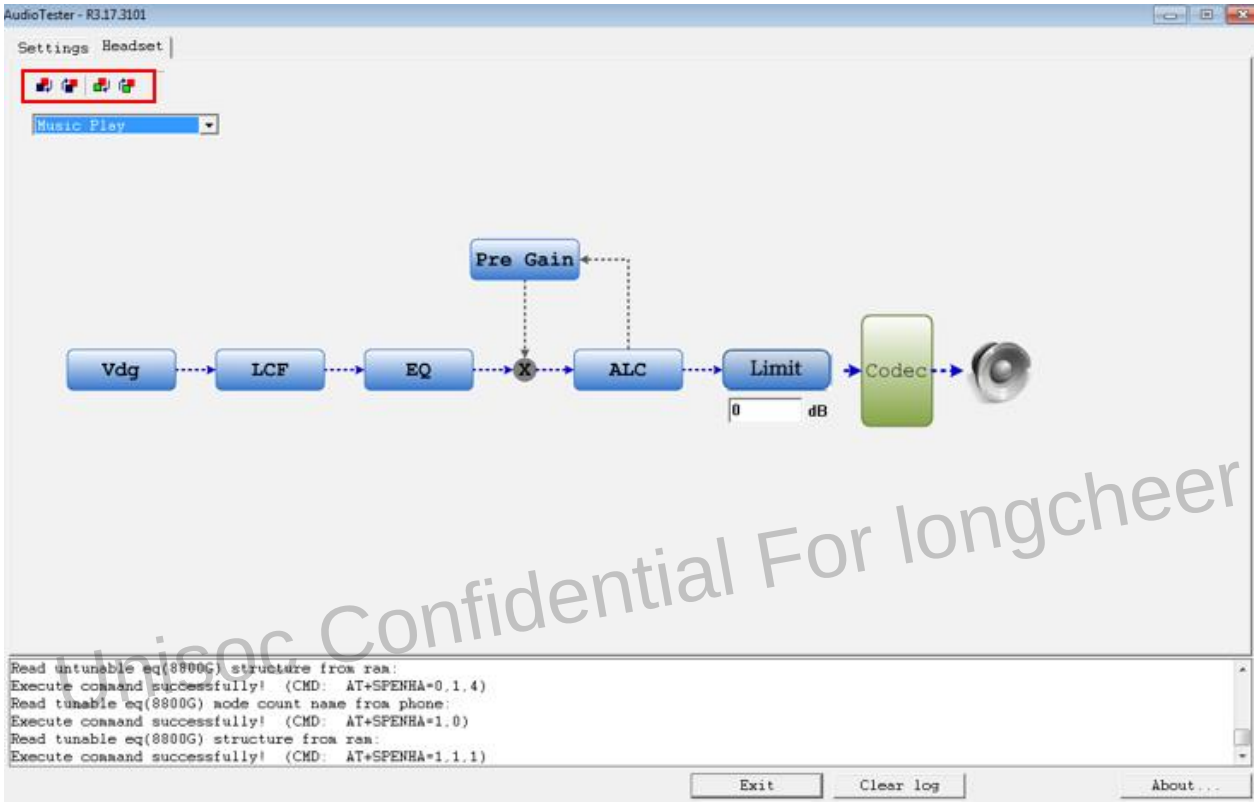

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Table 3-1 Icon and function description

Icon	Function	Description
	Write parameters to RAM	Write parameters of current mode to DUT RAM.
	Read parameters from RAM	Read parameters from DUT RAM.
	Write parameters to Flash	Write parameters of current mode to DUT Flash.
	Read parameters from Flash	Read parameters from DUT Flash.

NOTE

The parameter will be restored to the value before writing if the power is off while writing in the parameter.

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3.3 Music Play tuning

3.3.1 Playback

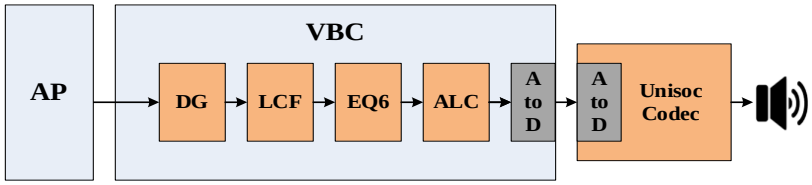
Playback contains six scenes, among which ISCHandsfree and ISCHeadfree belong to Smart-AMP scene.

- Handset
- Handsfree
- Headset
- Headfree
- ISCHandsfree
- ISCHeadfree

DG, LCF, EQ, ALC in VBC require Bypass or Off when tuning Audio effect through Smart-AMP. In general, only Vdg and Codec require to be tuned. For tuning methods, see **3.3.2 VBC_Vdg** and **3.3.7 Codec**.

Playback data block diagram includes tuning modules such as DG, LCF, EQ6 and ALC, as shown in **Figure 3-3**.

Figure 3-3 Playback data block diagram



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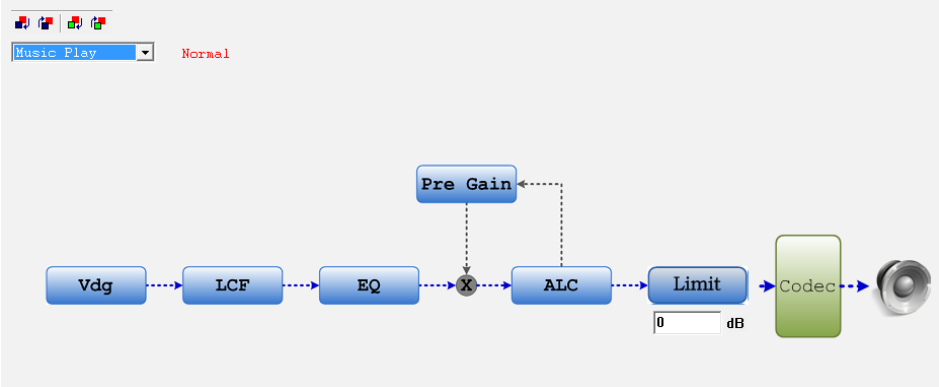
Tuning interface of Playback is shown in **Figure 3-4**.

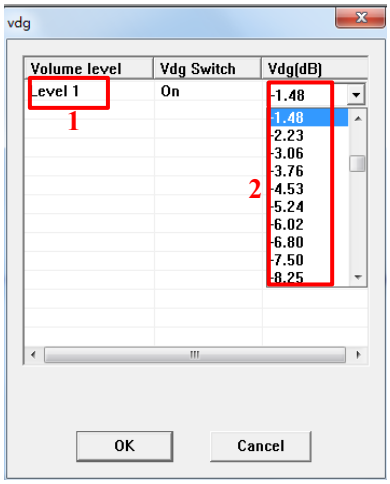
Figure 3-4 Playback tuning interface



3.3.2 VBC_Vdg

The module interface of VBC_Vdg is shown in **Figure 3-5**.

Figure 3-5 VBC_Vdg module interface



Red box 1: indicates level 1 volume.

Red box 2: clicks the drop-down box to select the appropriate gain.

3.3.3 LCF

LCF module is a Band in EQ6. If EQ6 is turned off, LCF will also be turned off. The interface of the module is as shown in **Figure 3-6**, and the description of the options on the interface is in **Table 3-2**.

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Figure 3-6 Interface of the LCF module

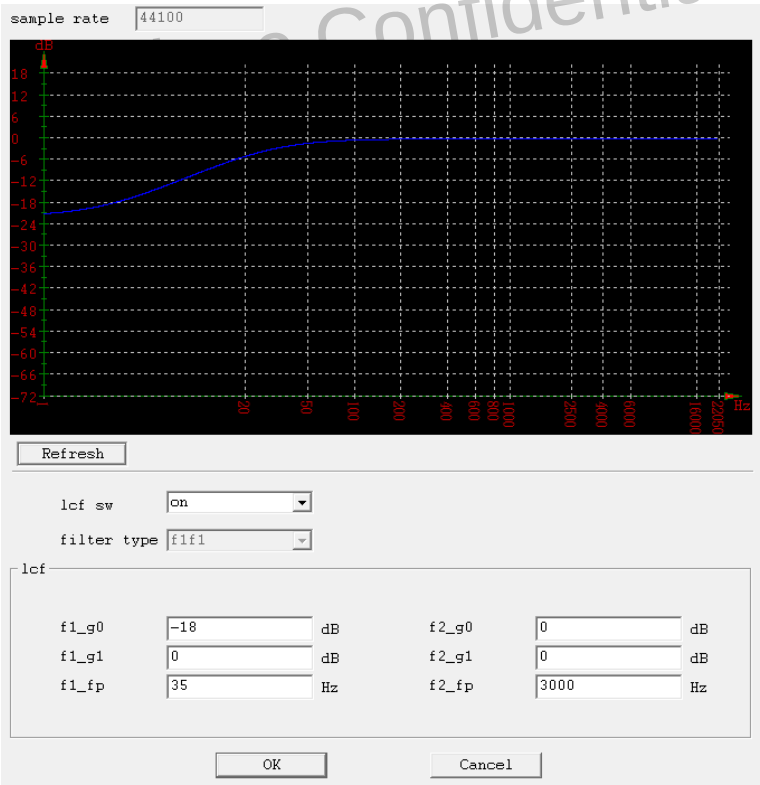


Table 3-2 Descriptions of options on the interface of the LCF module

Option	Description
sample rate	The sampling rate is set to 44.1KHz and the user cannot modified.
Refresh	Refresh the LCF parameter settings, and the corresponding curve is shown in **Figure 3-6**.
filter type	The user cannot modify filter type.
lcf	• f1_g0: the gain of the first filter at 0Hz. • f1_g1: the gain of the first filter at 22,050Hz. • f1_fp: the shear frequency of the first filter at -3dB. • f2_g0: the gain of the second filter at 0Hz. • f2_g1: the gain of the second filter at 22,050Hz. • f2_fp: the cut-off frequency of the second filter at -3dB.

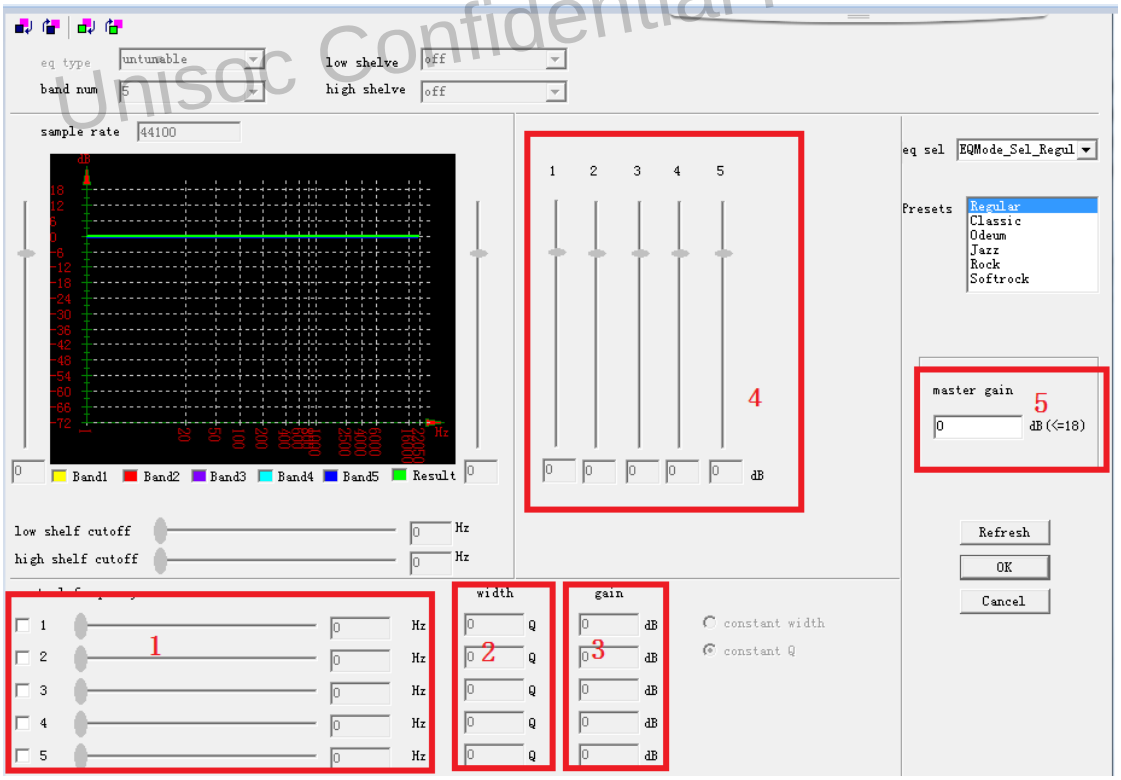
NOTE

Use the first filter to adjust the amplitude response at low frequencies, and use the second filter to adjust the amplitude response at high frequencies.

3.3.4 EQ

The interface of EQ module is shown in **Figure 3-7**.

Figure 3-7 EQ module interface



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EQ type consists of following two modes:

- untunable: Users cannot edit each mode of EQ in this mode, and band num, low shelve, and high shelve are not adjustable.
- tunable: Users can edit each mode of EQ in this mode in MMI interface.

CAUTION

Click the Refresh button in the lower right corner when you adjusted the EQ parameters, and then click button in the upper left corner to save.

EQ parameter descriptions is shown in **Table 3-3**.

Table 3-3 EQ parameter description

No.	Parameter	Description
1	Center frequency (f0)	Modify the center frequency of a frequency band by pulling the ball or modifying the value.
2	Width (Q)	Change the Q value of a frequency band. The definition of Q value is f0/△f (△f indicates the band width).
3	gain	Change the base gain of some frequency bands.
4	boost and band	boost: Modify the center frequency boost value of a frequency band by pulling the ball or modifying the value. band: • If the checkbox is marked as ☑, the corresponding frequency band is set to on. • If the checkbox is marked as ☐, the corresponding frequency band is set to off.
5	master gain	The overall gain of EQ.

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3.3.5 ALC

The default parameters of the ALC module interface are shown in **Figure 3-8**, and the interface descriptions are shown in **Table 3-4**.

Figure 3-8 ALC module interface

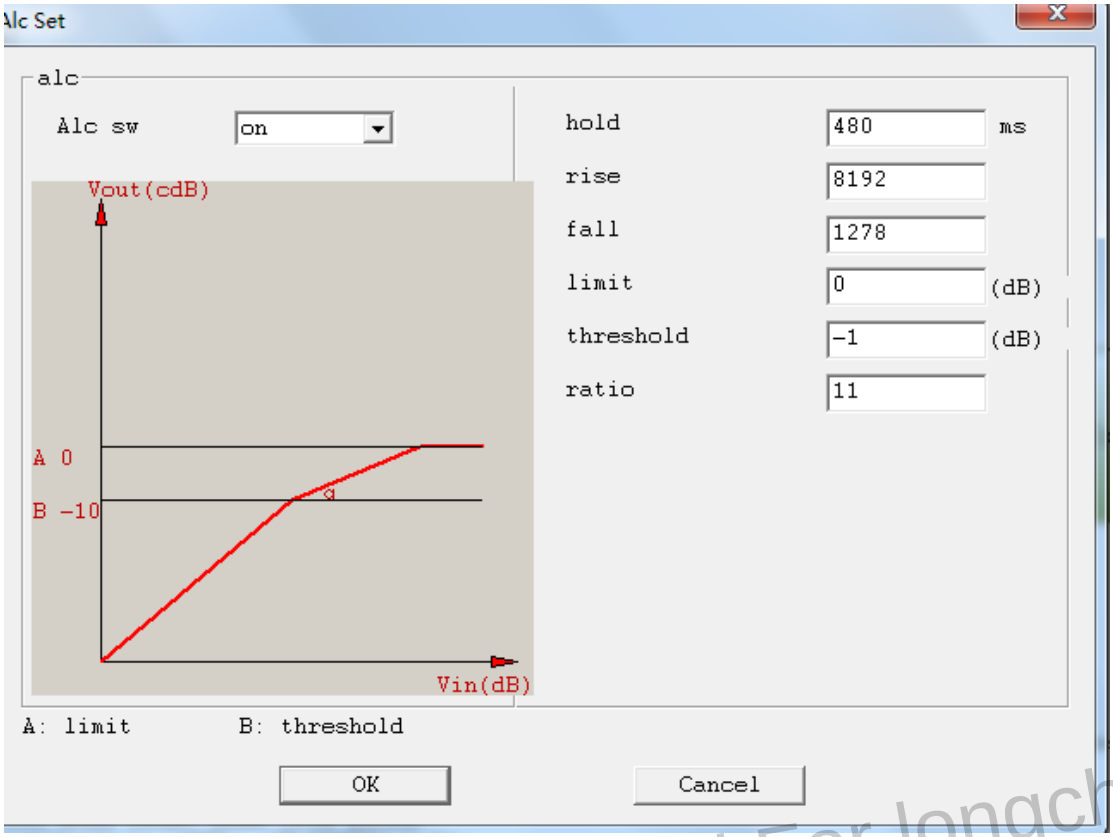

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Table 3-4 ALC interface description

Option	Description
Alc sw	Automatic volume control switch.
hold	The gain holding time before starting to release. It is recommended to use the default value.
rise	The tracking speed when the peak value rises. The larger the value, the faster the peak tracking speed. It is recommended to use the default value.
fall	The tracking speed when the peak value falls. The larger the value, the faster the peak tracking speed. It is recommended to use the default value.
limit	The limit level (dB). The larger the value, the faster the peak tracking speed.
threshold	The threshold level (dB), the maximum input signal level that is not affected by compression (limiting). And the input signals above the threshold level will be compressed.
ratio	The compression ratio. A measure of gain reduction when the input signal exceeds the threshold level. The definition of the compression ratio is output level increase/input level increase.

3.3.6 Inner PA

The interface of Inner PA is shown in **Figure 3-9**, and the interface descriptions are shown in **Table 3-5**.

Figure 3-9 Inner PA interface

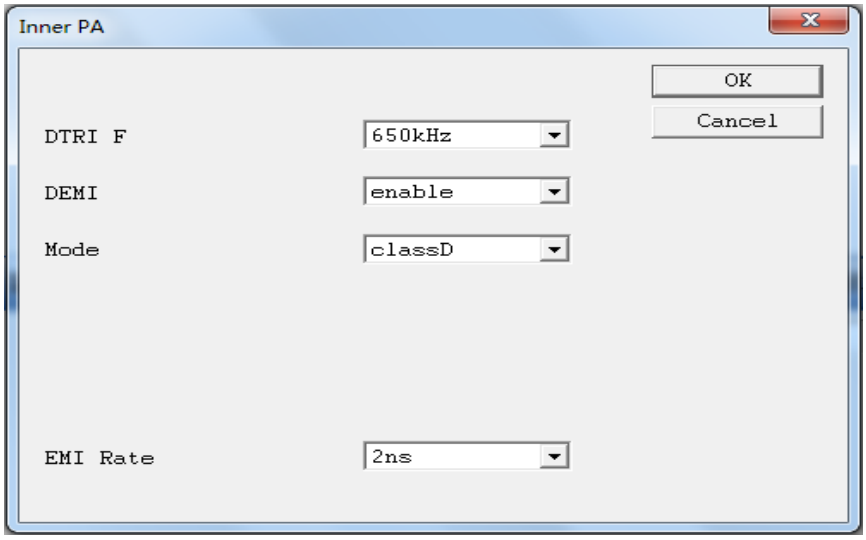

Table 3-5 Inner PA interface description

Option	Description
DTRI F	classD switching frequency.
DEMI	Enable or disable the EMI function.
Mode	Select PA mode (classAB or classD).
EMI Rate	The default EMI rate is 2ns. The frequency and energy of EMI radiation of classD are different in different configurations, and EMI rate can be tuned when FM/GPS/RF is receiving interference.

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3.3.7 Codec

Codec can set the analog gain and DAC gain, and its interface is shown in **Figure 3-10**.

- The Headset channel corresponds to HP_G.
- The Handset channel corresponds to EAR_G.
- The Handsfree channel corresponds to PA_G.

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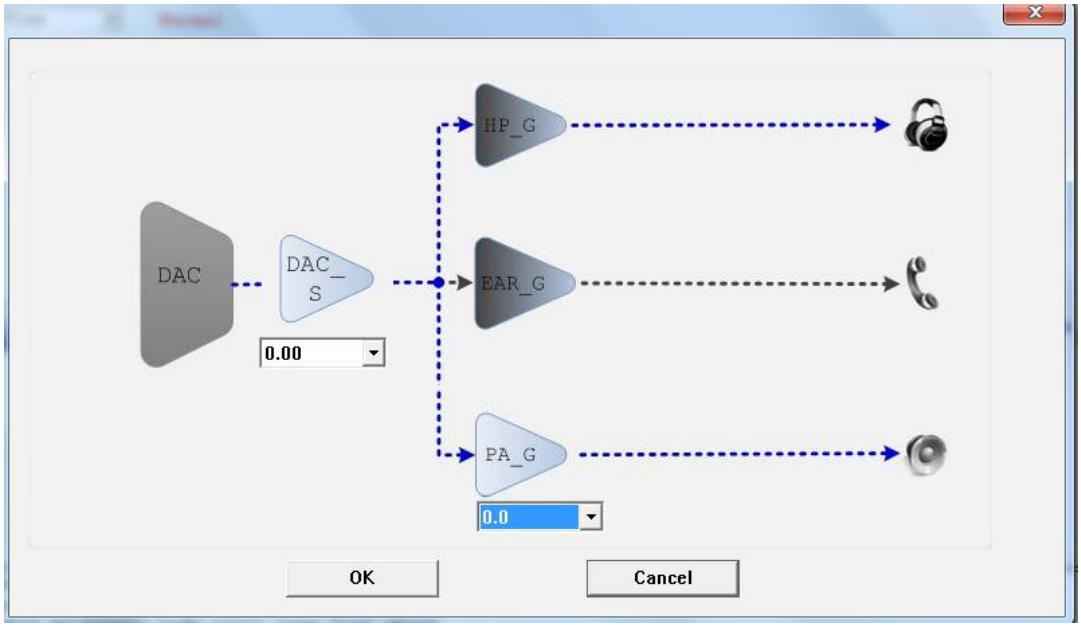
Figure 3-10 Codec interface



3.4 FM & FM Record tuning

3.4.1 FM & FM Record

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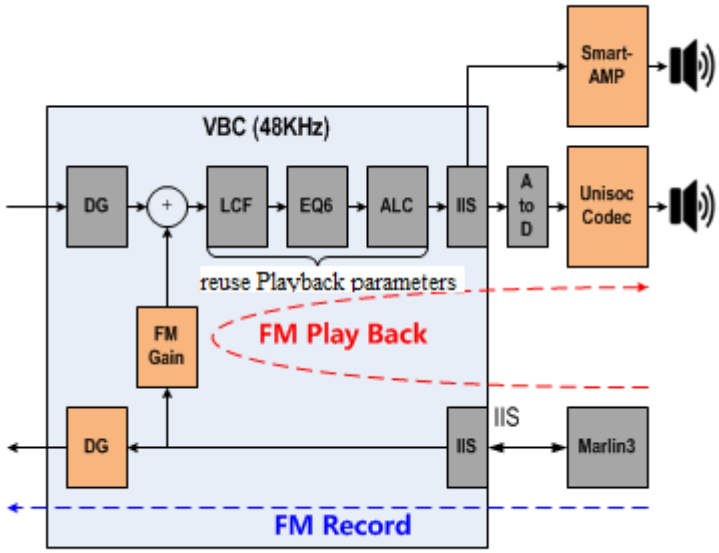
FM consists of three modes, and the data block diagram of FM is shown in **Figure 3-11**.

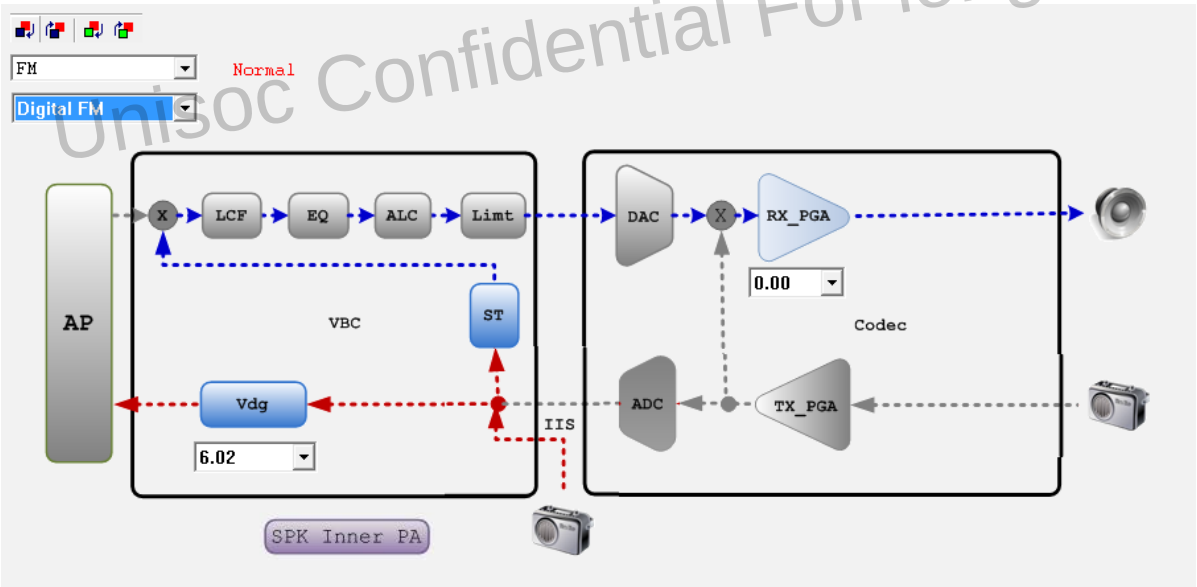
- Analog FM
- Analog FM VBC
- Digital FM

Figure 3-11 Data block diagram of FM



The data block diagram of FM contains VDG, ST and RX_PGA, as shown in **Figure 3-12**.

Figure 3-12 FM tuning interface



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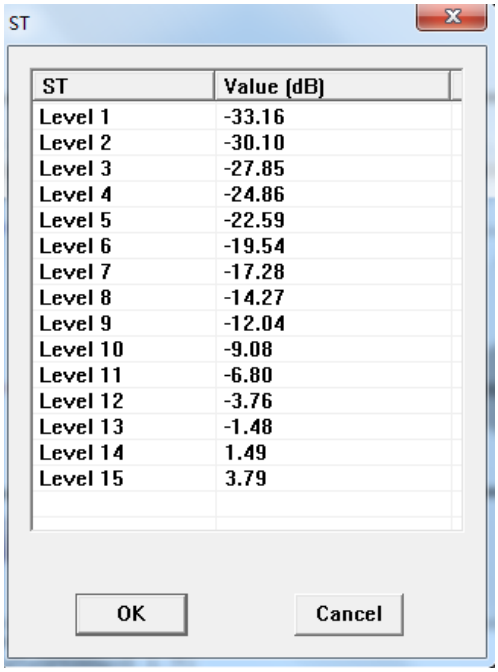
3.4.2 VBC_FM_Gain

The 15-level volume control gain for FM playback is shown in **Figure 3-13**.

Figure 3-13 ST interface



3.4.3 FM Rx PGA & SPK Inner PA

Codec parameter configuration of FM playback is shown in **Figure 3-14**, and the parameter configuration of Speaker PA is shown in **Figure 3-15.**

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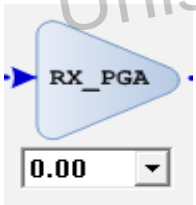
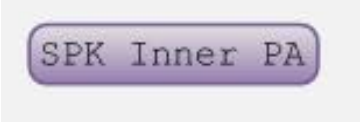
Figure 3-14 PGA gain in Handsfree/Headset channel



The PGA gain (Music/Headset/FM/RX_PGA/HP_G) setting value in the Headset channel is consistent with the parameters (Music/Headset/Playback/Codec/HP_G).

Figure 3-15 Speaker PA parameter configuration



The parameters of Music/Handsfree/FM/RX_PGA & SPK Inner PA and Music/Handsfree/Playback/Codec/PA_G & SPK Inner PA are the same.

NOTE

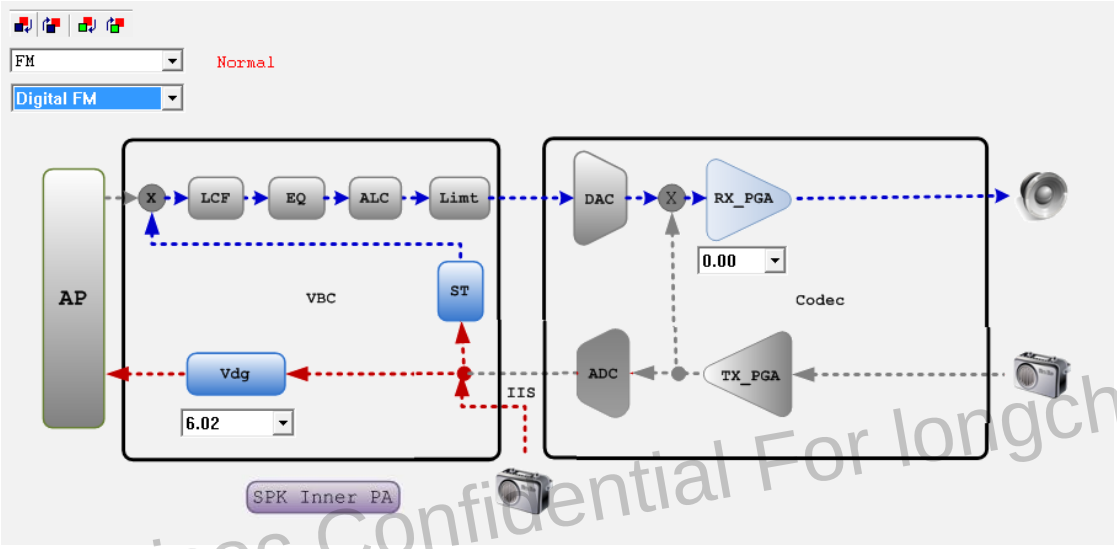
With differentiated hardware design, such as chip HPL output connecting to an external AMP to set the speaker as a speaker channel, you need to consult UNISOC engineers to modify corresponding parameter, and the tool will automatically configure to HP_G.

With differentiated hardware design, such as chip IIS output connecting to an external Smart-AMP to set the speaker as a speaker channel. You need to consult UNIOSC software staff to modify the corresponding parameter, and the tool will automatically configure the parameter and gray out part of codec block diagram.

3.4.4 VBC_Vdg (FM Record Gain)

FM Record level 1 volume control gain interface is shown in **Figure 3-16.**

Figure 3-16 FM Record interface



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3.5 Record & Unprocess Record

- Record: which is used for the upper-layer applications to deliver sound recording processed by sound effect algorithms.
- Un-process Record: which is used for the upper-layer applications to deliver sound recording processed by no sound effect algorithms. In this scene, turning on EQ to correct the frequency response curve problem caused by MIC and sound cavity is recommended. NR and ALC is recommended to turn off.

The data block diagram of Record & Unprocess Record is shown in **Figure 3-17**. The tuning interface of Handsfree Record & Unprocess Record is shown in **Figure 3-18**.

Figure 3-17 Data block diagram of Record & Unprocess Record

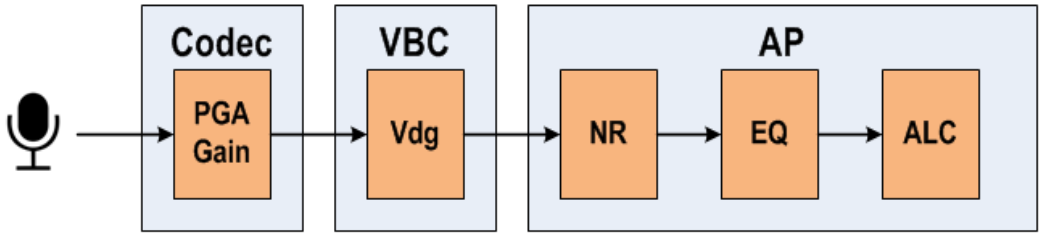
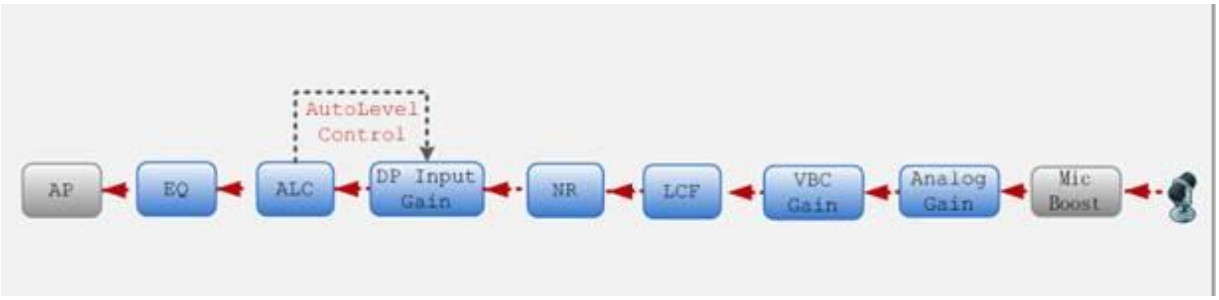


Figure 3-18 Tuning interface of Handsfree Record & Unprocess Record



3.5.1 Analog gain

- Level 1 volume.
- Click the drop-down box and select the appropriate gain.

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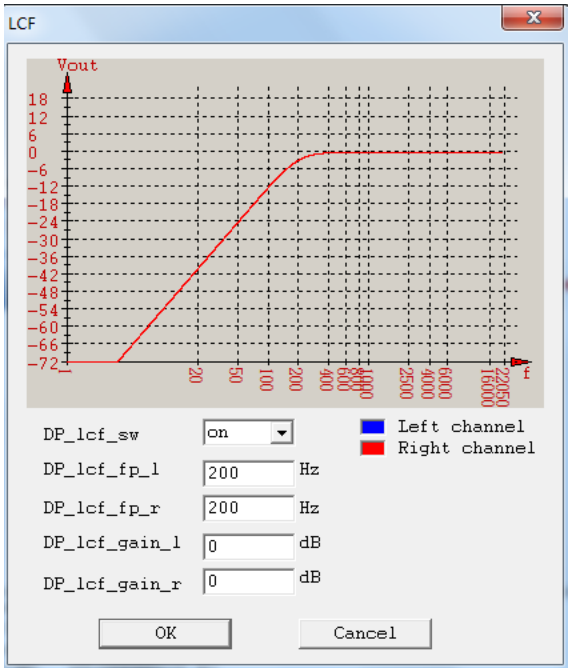
3.5.2 VBC gain

- You can choose switch On or Off.
- Click the drop-down box and select the appropriate gain.

3.5.3 LCF

The interface of LCF is shown in **Figure 3-19**.

Figure 3-19 LCF interface



- DP_lcf_sw: open switch On/Off.
- DP_lcf_fp_l/r: two high pass filters, which can be filled with cut-off frequency. It is recommended to set both filters to the same value.
- DP_lcf_gain_l/r: value of the lcf gain.

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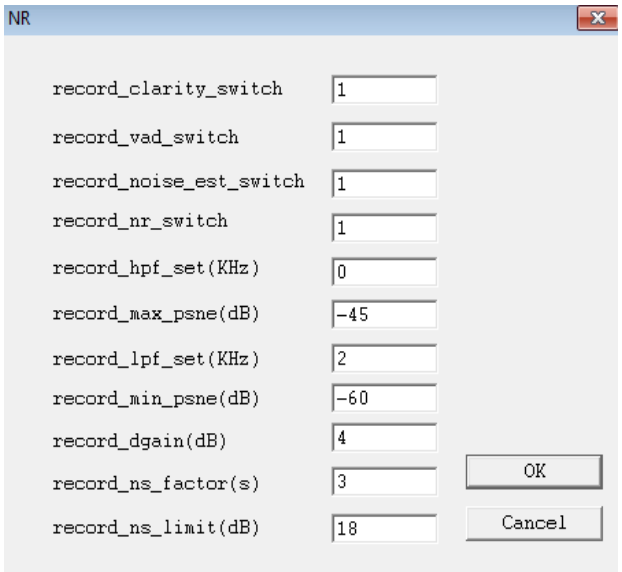
3.5.4 NR

The interface of NR is shown in **Figure 3-20** .

Figure 3-20 NR interface



- record_clarity_switch:
  - 1: On
  - 0: Off
- record_vad_switch: voice detection (default enabled).
- record_noise_est_switch: noise estimation switch (default enabled).
- record_nr_switch: noise suppression switch (default enabled).
- record_min_psne: minimum noise threshold
  - If signal is less than min_psne, output 0.
  - If signal is greater than min_psne, noise is suppressed according to ns_limit.
- record_max_psne: maximum noise threshold.
  - If signal is less than min_psne, output unchanged.
  - If signal is greater than min_psne, noise is suppressed according to ns_limit.
- record_hpf_set: high pass filter cut-off frequency.
- record_lpf_set: low pass filter cut-off frequency, which value is less than 24,000Hz.
- record_dgain: noise cancellation digital gain.
- record_ns_factor: noise suppression tracking step (default value is 1).
  - The larger the value, the faster the noise tracking.
  - If the value is too large, the sound will be unstable.
  - The smaller the value, the slower the noise tracking.
  - If the value is too small, noise suppression will be poor.
- record_ns_limit: noise suppression value (dB).

3.5.5 ALC

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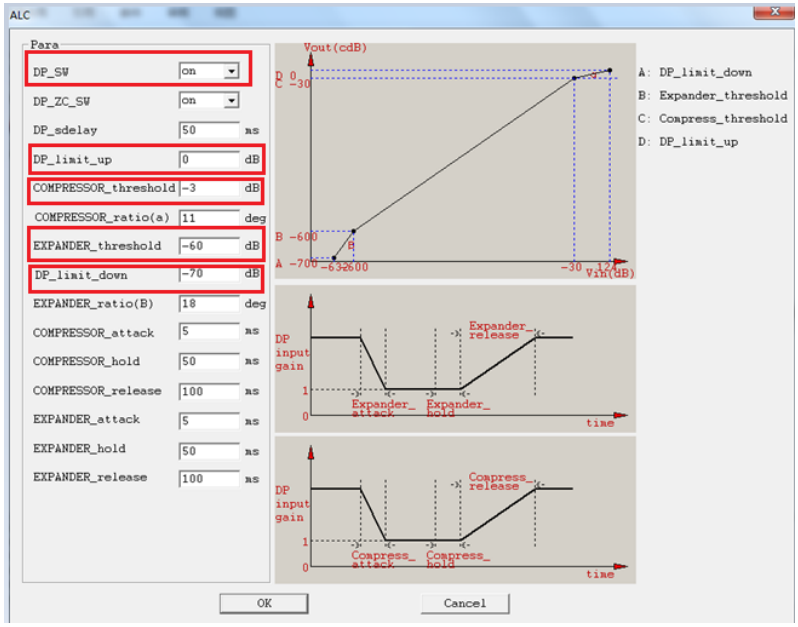
The parameters not marked in the red box in **Figure 3-21** are set to default values.

Figure 3-21 ALC interface



- DP_SW: ALC switch On/Off.
- DP_ZC_SW: Zero Cross function, the gain value is changed after the zero point. If DP_SW is turned on, ALC will be turned on, and it does not affect whether DP_ZC_SW is On or Off.
- DP_sdelay: digital processing debounce delay.
- DP_limit_up: the maximum output level of the compression limit level.
- COMPRESS_threshold: the compression threshold level, the maximum input signal level that is not affected by compression (limiting). And the input signals above the threshold level will be compressed.
- COMPRESS_ratio(a): a measure of gain reduction when the input signal exceeds the threshold level. Defined as output level increase/input level increase.
- EXPANDER_threshold: the extended threshold level, the maximum input signal level that is not affected by extending (limiting). And the input signals below the threshold level will be extended.
- DP_limit_down: the minimum output level of the extended threshold level.
- EXPANDER_ratio(B): a measure of gain reduction when the input signal below the threshold level. Defined as output level increase/input level increase.
- COMPRESSOR_attack: the compression response time, which indicates the time between the signal exceeds the threshold level and the compression control starts.
- COMPRESSOR_hold: the compression hold time, which indicates the time that the gain is held before the release.
- COMPRESSOR_release: the compression release time, which indicates the time between the end of the overload signal and the elimination of the compression control.
- EXPANDER_attack: the extended response time, which indicates the time between the signal below the threshold level and the extending control starts.
- EXPANDER_hold: the extended hold time, which indicates the time that the gain is held before the release.
- EXPANDER_release: the extended release time, which indicates the time between the end of the overload signal and the elimination of the extended control.

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3.5.6 EQ

The tuning method see **3.3.4 EQ**.

4 Parameters submission

The .nvm file and the .xml file are required to integrate into the software version when submitting the parameters. To convert the .nv file into the .nvm or the .xml before the integration, you need to prepare the following tools.

- NVEditor: applicable to .nvm structure projects.
- NVTool: applicable to .xml structure projects.

4.1 Export NV file

AudioTester exports NV files for offline viewing when there is no mobile phone.

4.2 Import NV file

NV is an offline file of audio parameters that saved locally. When there is no mobile phone, you can import NV for viewing and editing through AudioTester.

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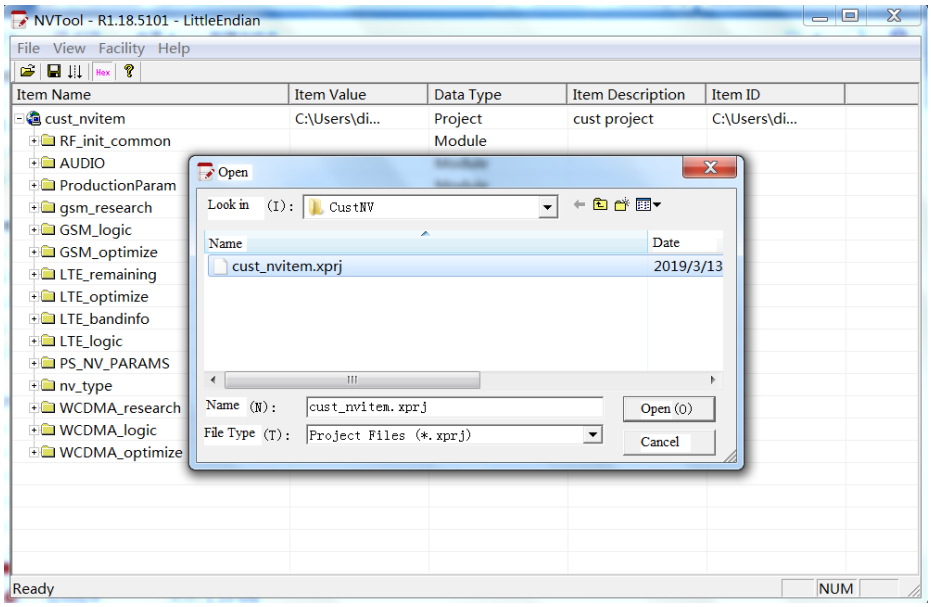
4.3 NVTool generates XML file

The steps to generate the XML files are as follow.

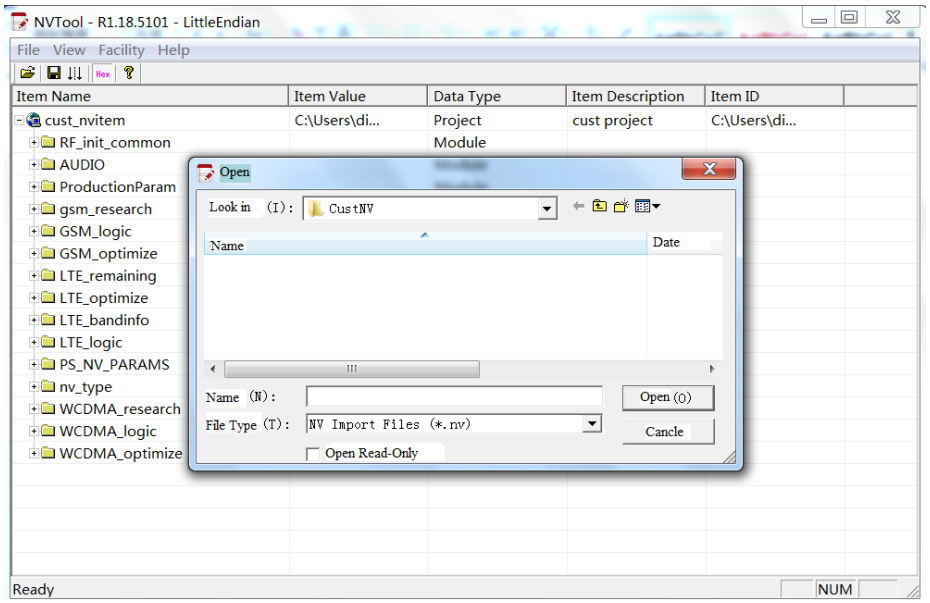
Step 1 Open the NVTool, and click **File → OpenProject** in the top left corner to open the .xprj file as shown in **Figure 4-1**.

Figure 4-1 Open .xprj file through NVTool



Step 2 Select import function in the drop-down box of the facility option as shown in **Figure 4-2**.

Figure 4-2 NVTool import NV operate interface



Step 3 Select the modified NV file to import.

After importing the modified NV file, the modified content will be marked in red, as shown in **Figure 4-3**.

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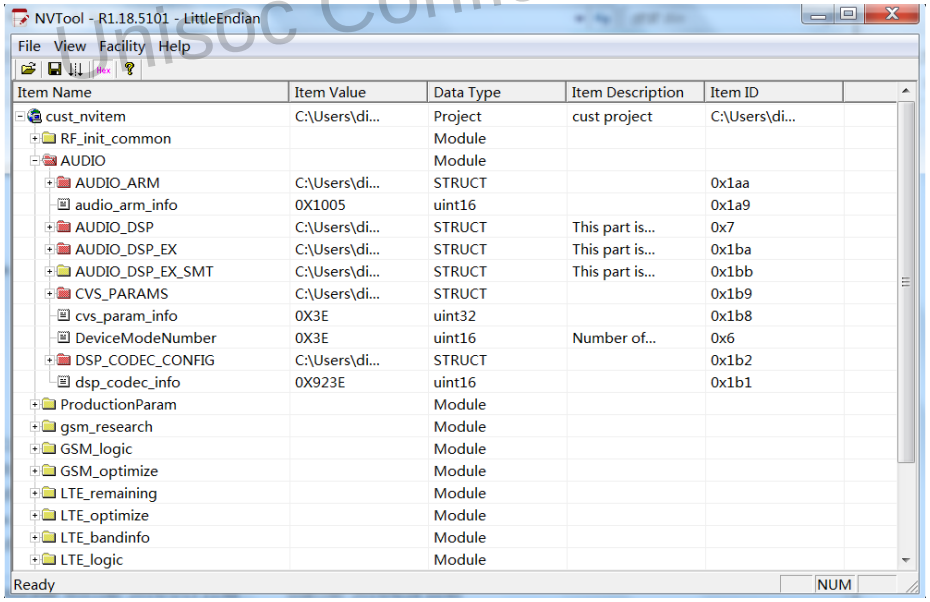
Figure 4-3 Interface after importing the NV file



Step 4 Click **File → save image → save project** to save the project (.xml) file.

----End

Compare the saved .xml with the original .xml file to check the accuracy of the modified content. Upload the generated .xml file to the corresponding branch or forward it to the software staff.

4.4 NVEditor generates NVM file

To generate the NVM file, see **4.3 NVTool generates XML file**.

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5 FAQ

5.1 Optimize Call Delay Fail

Step 1 Select the parameters of the mode to be optimized such as handset, handsetWB, handsetLTENB.

Step 2 Click **CVS** button in the AudioTester tuning interface to enter CVS tuning interface.

Step 3 Click the **Channel Mode** button of Rx Path to expand the drop-down options.

Step 4 Select **Copy** (Bypass CVS Rx Path), and reduce the Delay by about 26ms.

----End

The Rx un-bypass before Call Delay optimization is shown in **Figure 5-1**, and the optimized Rx un-bypass is shown in **Figure 5-2**.

Figure 5-1 Rx un-bypass before Call Delay optimization

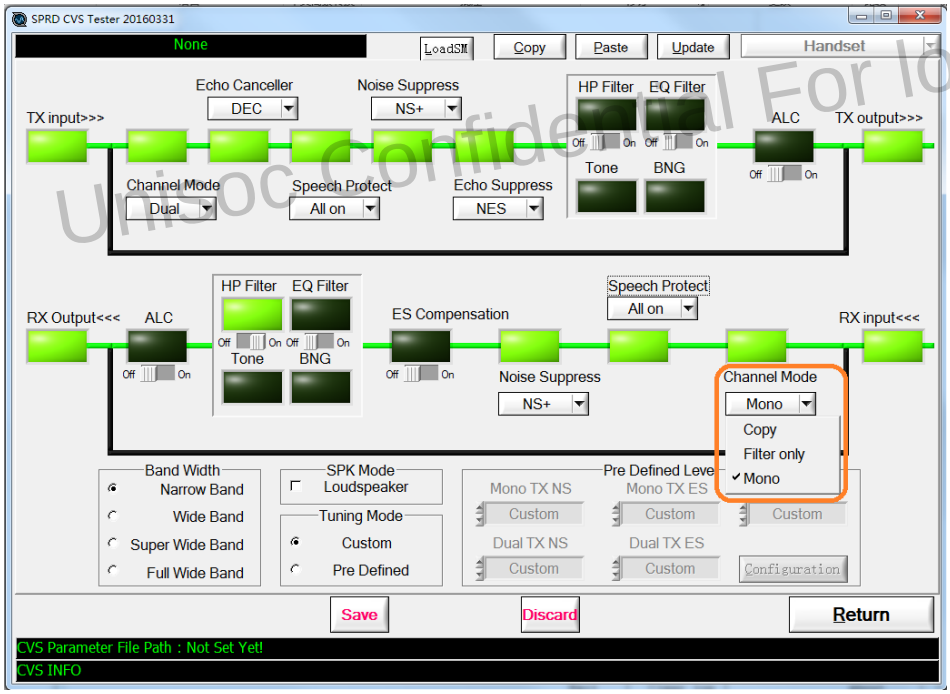

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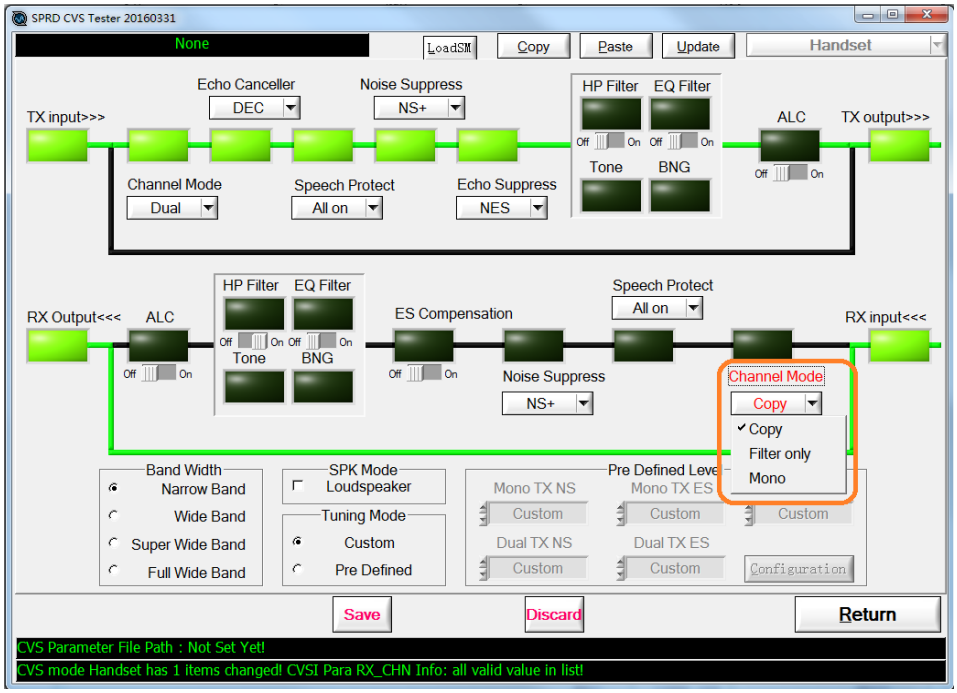
Figure 5-2 Rx un-bypass after Call Delay optimization



The Handset parameter modification is shown in **Table 5-1**. The HandsetWB parameter modification is shown in **Table 5-2**.

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Table 5-1 Handset parameter modification

Parameter address	Call Delay value before optimization	Call Delay value after optimization	Description
audio\audio\Handset\AudioStructure\extend\extend[83]	0x0	0x1	Bit[0]: 0: DSP uplink NR off 1: DSP downlink NR on
audio\audio\Handset\AudioStructure\vbc\IIS_16K_ctrl0	0x0	0x1	Bit[0]: 0: CVS uplink un-bypass 1: DSP downlink bypass
cvs_params\cvs_params\Handset\cvs_ctrl_params\dl_AEC	0x5e00	0x1000	CVS Echo delay time 0x1000 = 8ms 0x5e00 = 47ms
lvve_params\lvve_params\Handset\lvve_ctrl_params\ControlParams_Tx\BulkDelay	0x17c	0x78	NXP Echo delay time 0x17c = 380 samples 0x78 = 120 samples

Parameter address	Call Delay value before optimization	Call Delay value after optimization	Description
AUDIO\CVS_PARAMS\Handset\cvs_ctrl_params\RX_MODUAL\CHN	0x1	0xffff	CVS Rx Path Setting 0x1: Mono 0xffff: Copy

Table 5-2 HandsetWB parameter modification

Parameter address	Call Delay value before optimization	Call Delay value after optimization	Description
audio\audio\Handset\AudioStructure\extend\extend[83]	0x0	0x1	Bit[0]: 0: DSP uplink NR off 1: DSP downlink NR on
audio\audio\Handset\AudioStructure\vbc\IIS_16K_ctrl0	0x0	0x1	Bit[0]: 0: CVS uplink un-bypass 1: DSP downlink bypass
cvs_params\cvs_params\Handset\cvs_ctrl_params\dl_AEC	0x5600	0x2400	CVS Echo delay time 0x2400 = 18ms 0x5600 = 43ms
lvve_params\lvve_params\Handset\lvve_ctrl_params\ControlParams_Tx\BulkDelay	0x2f8	0xf0	NXP Echo delay time 0x2f8 = 760 samples 0xf0 = 240 samples
AUDIO\CVS_PARAMS\Handset\cvs_ctrl_params\RX_MODUAL\CHN	0x1	0xffff	CVS Rx Path Setting 0x1: Mono 0xffff: Copy

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5.2 Quick tuning CVS echo noise

Step 1 Select the parameters of the mode to be optimized such as handset, handsfree, handsetWB, handsfreeLTENB.

Step 2 Click **CVS** button in the AudioTester tuning interface to enter CVS tuning interface.

Step 3 Select **Pre Defined** in Tuning Mode.

Step 4 Select inhibitory intensity of each module in red box 4, as shown in **Figure 5-3**.

Figure 5-3 CVS echo noise quick tuning

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- Mono TX NS: single MIC uplink noise Pre Defined Level setting.
  - Custom: customer customization
  - 1: the weakest noise suppression
  - 6: the strongest noise suppression
- Mono TX ES: single MIC uplink echo noise Pre Defined Level setting.
  - Custom: customer customization
  - 1: the weakest noise suppression
  - 6: the strongest noise suppression
- Mono RX NS: downlink noise Pre Defined Level setting.
  - Custom: customer customization
  - 1: the weakest noise suppression
  - 6: the strongest noise suppression
- Dual TX NS: dual MIC uplink noise Pre Defined Level setting.
  - Custom: customer customization
  - 1: the weakest noise suppression
  - 6: the strongest noise suppression
- Dual TX ES: dual MIC uplink echo noise Pre Defined Level setting.
  - Custom: customer customization
  - 1: the weakest noise suppression
  - 6: the strongest noise suppression

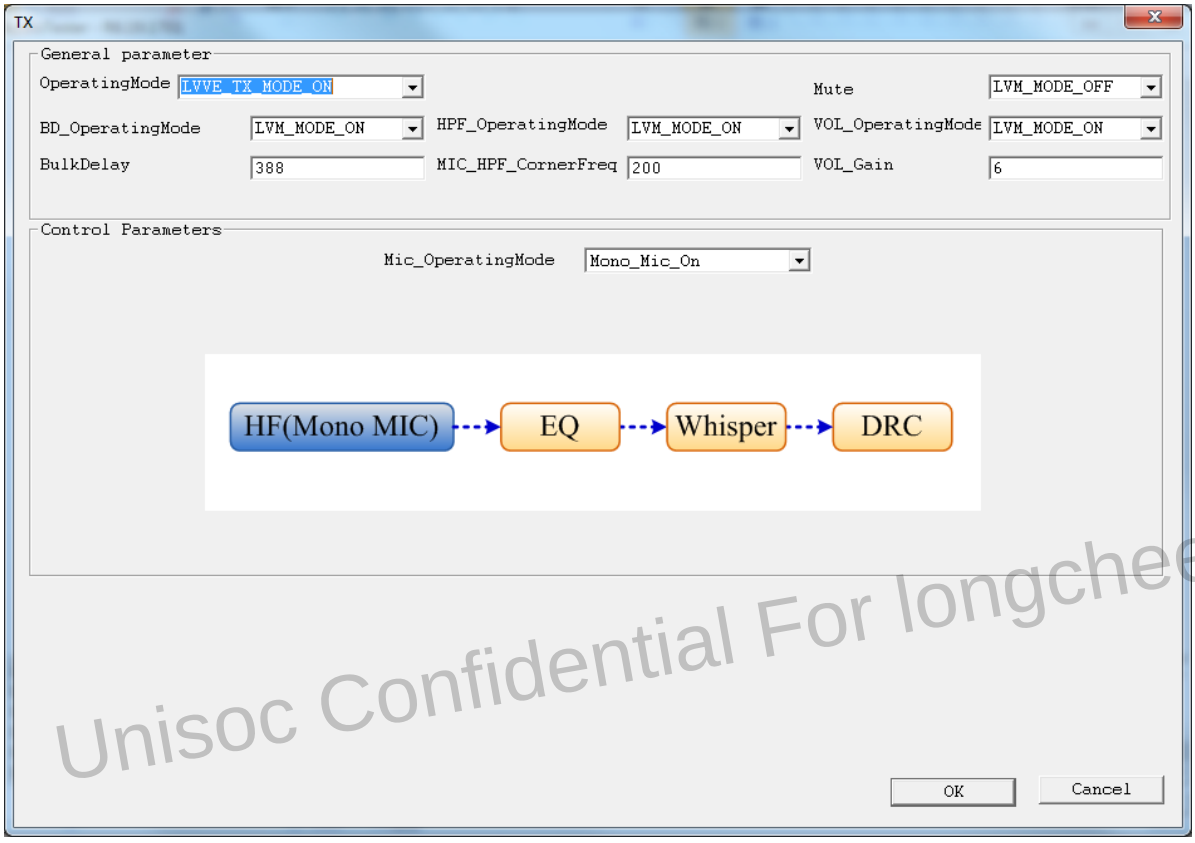
----End

5.3 NXP tuning

5.3.1 NXP TX

The tuning interface of NXP TX is shown in **Figure 5-4**.

Figure 5-4 NXP TX tuning interface



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- OperatingMode: LVVE_RC_MODE_ON (Default parameters)
- Mute: LVM_MODE_OFF (Default parameters)
- BD_OperatingMode: LVM_MODE_ON, BulkDelay module is ON.
- HPF_OperatingMode: LVM_MODE_ON, high pass filter module is ON.
- VOL_OparatingMode: LVM_MODE_ON, volume control module is ON.
- BulkDelay: echo noise delay
  - in NB mode: 380 samplings
  - in WB mode: 760 samplings
  - in NB mode: Call Delay optimization, NXP Rx Bypass: 120 samplings.
  - in WB mode: Call Delay optimization, NXP Rx Bypass: 240 samplings.
- MIC_HPF_CornerFreq: 200Hz (MIC uplink high pass filter frequency)
- VOL_Gain: 6Db (MIC uplink gain)
- Mic_OperatingMode:
  - Mono_Mic_On, single MIC

- Dual_Mic_On, dual MIC

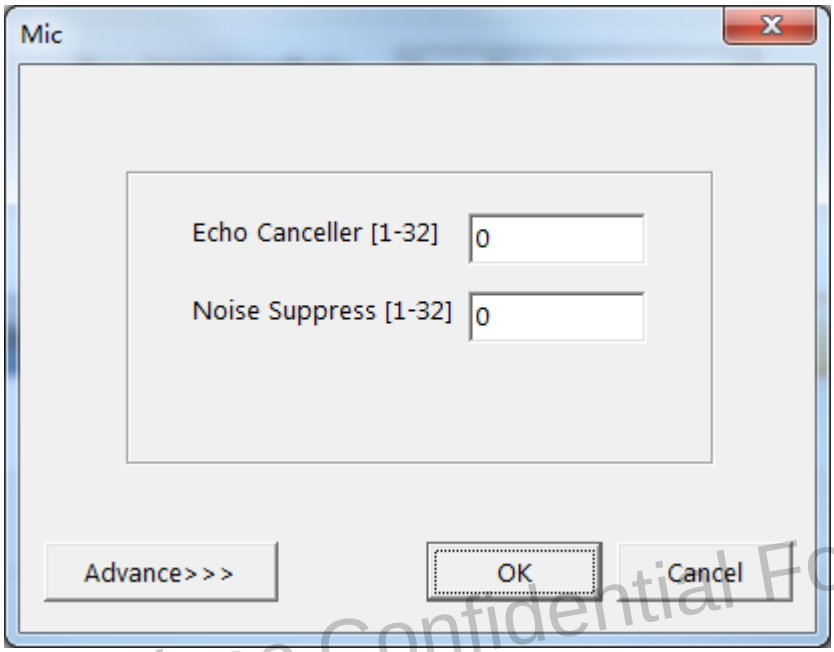
NOTE

The NXP library consists of single MIC and double MIC. Audio software is required to confirm the NXP library to be used.

5.3.2 NXP AEC NS quick tuning

The quick tuning of NXP AEC NS is shown in **Figure 5-5**.

Figure 5-5 NXP AEC NS quick tuning interface



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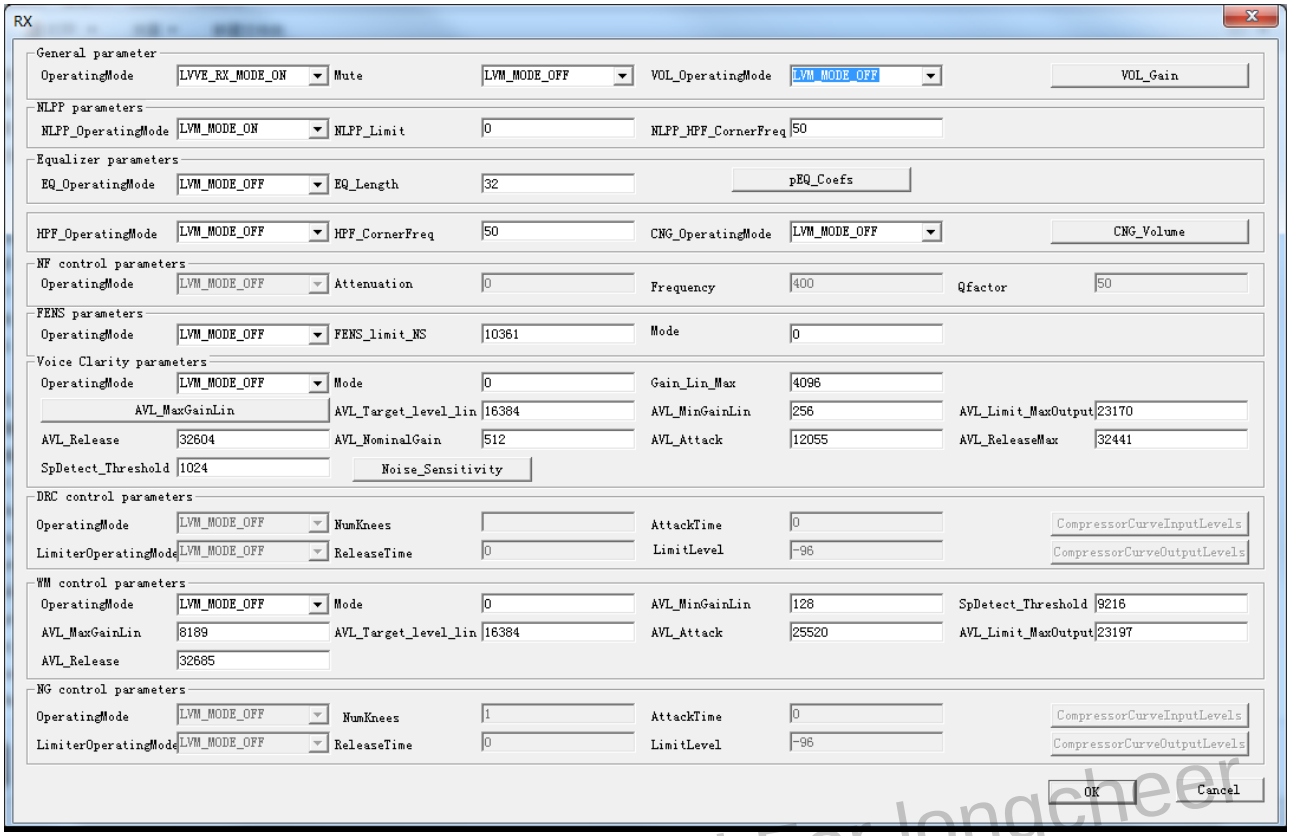
- Echo Canceller: the echo noise suppression degree which is ascending along with the numerical value.
  - 1: weak echo suppression
  - 32: strong echo suppression
- Noise Suppress: the noise suppression degree which is ascending along with the numerical value.
  - 1: weak echo suppression
  - 32: strong echo suppression

5.3.3 NXP Rx

NXP Rx interface is shown in **Figure 5-6**.

Figure 5-6 NXP Rx interface



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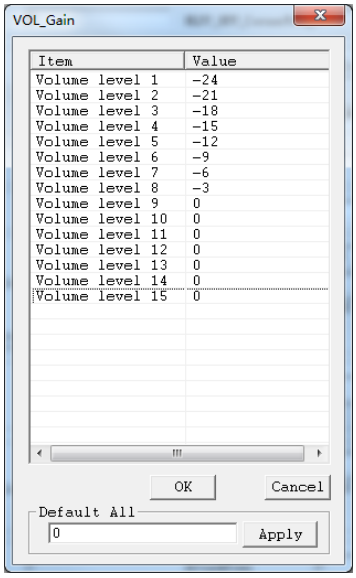
- OperatingMode: LVVE_RC_MODE_ON (default parameter)
- Mute: LVM_MODE_OFF (default parameter)
- VOL_OperatingMode:
  - LVM_MODE_OFF (NXP Volume Gain is invalid)
  - LVM_MODE_ON (NXP Volume Gain is valid, and it uses parameters in Vol_Gain as shown in **Figure 5-7**.)
- Vol_Gain: the recommended setting at level 9 volume is shown in **Figure 5-7**.

Figure 5-7 Vol_Gain



- NLPP_OperatingMode: LVM_MODE_ON (default parameter)
- NLPP_Limit: non-linear suppression threshold
  - Generally set to 0dB.
  - When the downlink sound distortion is large, the threshold can be appropriately lowered, such as -3dB, -6dB, but it will cause the volume to decrease.
- NLPP_HPF_CornerFreq: non-linear suppression high pass filter corner frequency.
  - Handset: set to 50Hz.
  - Handsfree: set to 300Hz.
- EQ_OperatingMode: LVM_MODE_OFF (default parameter, which is not recommended.)
- HPF_OperatingMode: LVM_MODE_ON
- HPF_CornerFreq: high pass filter frequency setting
  - Handset: set to 50Hz.
  - Handsfree: set to 300Hz.
- CNG_OperatingMode: LVM_MODE_ON / LVM_MODE_OFF (Comfort noise On/Off)
- CNG_Volume: as shown in **Figure 5-8**.

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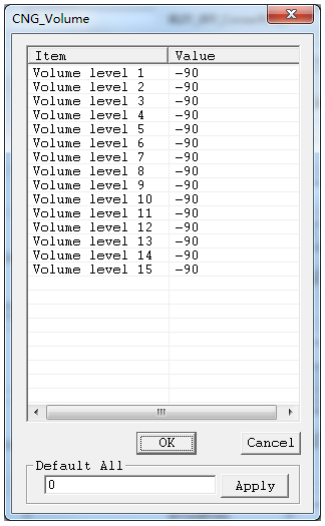
Figure 5-8 CNG_Volume



NOTE

Other modules are not recommended to be used.

5.4 Differential configuration of Loopback channel

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The loopback of the platform needs no Modem to compile and decode when the VoIP scene is used by default. Customer differentiation loopback consists of two categories:

- The scene where the platform audio channel is maintained, and the uplink and downlink are compiled and decoded by Modem.
- The scene of recording and playback.

The above scenes require software AE staff to provide differential packages to modify the channel, and hardware AE staff to debug parameters. When there are differentiated requirements, customers can submit bugs and transfer them to software AE staff and hardware AE staff through CPM.

5.5 Differential configuration of Audio channel

When there are differentiated requirements, customers can submit bugs and transfer them to software AE staff and hardware AE staff through CPM.

5.6 Call recording tuning method

1. Find the parameter of corresponding mode in NV.
2. Find the uplink recording gain (DSP_VOLUME[2]) and downlink recording gain (DSP_VOLUME[3]) during a call. The parameters are as shown in **Table 5-3**.

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Table 5-3 Call recording parameter description

Parameter	Definition	Description
DSP_VOLUME[2]	Uplink channel gain of a call recording.	0x1000 = 0dB 0x2000 = 6dB 0x0 (Parameter protection, default to 0dB)
DSP_VOLUME[3]	Downlink channel gain of a call recording.	0x1000 = 0dB 0x2000 = 6dB 0x0 (Parameter protection, default to 0dB)

Taking the modification of the parameters of the handset GSM mode as an example.

AUDIO\AUDIO_DSP\Handset\AudioStructure\dsp_volume\dsp_volume[2] = 0x2000 ;//6dB

AUDIO\AUDIO_DSP\Handset\AudioStructure\dsp_volume\dsp_volume[3] = 0x2000 ;//6dB

5.7 Change volume level for calls

1. Find the parameter of corresponding mode in NV.
2. Find DSP_VOLUME[0] and DSP_VOLUME[1]. The parameters are as shown in **Table 5-4**.

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Table 5-4 Parameter description

Parameter	Definition	Description
DSP_VOLUME[0]	The step length of each gain when the volume is expanded.	0xb53 = -3dB
DSP_VOLUME[1]	The maximum volume supported by DSP.	0x9 = 9 Level volume 0xF = 15 Level volume 0x0（Parameter protection, less than and default to 9 level）

Taking the modification of the parameters of the handset GSM mode as an example.

AUDIO\AUDIO_DSP\Handset\AudioStructure\dsp_volume\dsp_volume[0] = 0xb53 ;//-3dB

AUDIO\AUDIO_DSP\Handset\AudioStructure\dsp_volume\dsp_volume[3] = 0x9 ;//9 Level volume

5.8 Software algorithm configuration uplink TDD noise

1. Find the parameter of corresponding mode in NV.
2. Find extend[83], as shown in **Table 5-5**.

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Table 5-5 Software algorithm parameter configuration

Parameter	Definition	Description
extend[83]	Bit 0: downlink NR function switch.	0: Off 1: On
	Bit 8: uplink NR function switch.	0: Off 1: On
	Bit 12: uplink TDNC function switch.	0: Off 1: On

Taking the modification of the parameters of the handset GSM mode as an example.

AUDIO\AUDIO_DSP\Handset\AudioStructure\ extend\extend[83] = 0x101; TDNC Off

AUDIO\AUDIO_DSP\Handset\AudioStructure\ extend\extend[83] = 0x1101; TDNC On

5.9 Downlink limit output for calls

1. Find the parameter of corresponding mode in NV.
2. Find extend2[104], as shown in **Table 5-6**.

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Table 5-6 Modification of downlink limit output for calls

Parameter	Definition	Description
extend2[104]	Downlink output limit.	0x1000 = 0dB 0xa18 = -4dB

Taking the modification of the parameters of the handset GSM mode as an example.

AUDIO\AUDIO_DSP\Handset\AudioStructure\extend2\extend2[104] = 0x1000 ; 0dB

AUDIO\AUDIO_DSP\Handset\AudioStructure\extend2\extend2[104] = 0xa18 ; -4dB

5.10 Tuning handsfree downlink distortion

See **2.3.14** Dis HF.

5.11 Set the L/R channel signal of the call headset in reverse

1. Find the parameter of corresponding mode in NV.
2. Find dl_DA_device_internal as shown in **Table 5-7**.

Table 5-7 Headset L/R channel signal reverse setting

Parameters	Definition	Description
dl_DA_device_internal	Bit[13] : DA1 Data reverse (R) Bit[12]: DA0 Data reverse (L)	0: Reverse 1: Not reverse

5.12 IIS parameter configuration during calls

1. Find the parameter of corresponding mode in NV.
2. Find extend2[117] as shown in **Table 5-8**, and extend2[118] as shown in **Table 5-9**.

Table 5-8 iis_set0 (extend2[117]): 0X10f9

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	PCM_CYCLE				IIS_CLKD											

- Bit[11:0]: IIS_CLKD
  - If IIS/PCM is running in master mode, IIS_CLKis valid. IIS/PCM is used to specify the ratio between IIS_CLK and IIS_CLKD.
  - IIS_CLK = clk_iis/2*(IIS_CLKD+1)
  - clk_iis of SC9820E = 128M
- Bit[15:12]: PCM_CYCLE
  - Used as muti_slot.

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Table 5-9 iis_set1(extend2[118]): 0xa45

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	Reserved		pcm_slot			Trans from lsb	bb used chan	active level left channel	bits per chan		sck inverse	operate mode	sync mode		device sel	

- Bit[1:0]: device_sel
  - 00: Voice band enable
  - 01: PCM enable
  - 10: IIS enable
  - 11: VBC+IIS is used for external SmartPA projects, using VBC interrupt and reading data from IIS as echo reference signal.
- Bit[3:2]: sync_mode
  - 00: long frame mode in PCM mode; MSB is aligned in IIS mode.
  - 01: short frame mode in PCM mode; IIIS is aligned in IIS mode.
- Bit[4]: operate_mode

- 0: used as master mode.
- 1: used as slave mode.

- Bit[5]: sck_inverse
  - 0: enable
  - 1: disable
- Bit[7:6]: bits_per_channel
  - 00: 8Bits
  - 01: 16Bits
  - 10: 32bits
  - 11: invalid
- Bit[8]: active level left channel
  - 0: low level of left channel
  - 1: high level of left channel
- Bit[9]: bb used chan
  - 0: LSB，low 16 bits
  - 1: MSB，high 16 bits
- Bit[10]: trans from lsb
  - 0: low level of left channel
  - 1: high level of left channel
- Bit[11:13]: pcm_slot

Bluetooth mode:

BTHSD8K: extend2[117] = 0x10f9 ; extend2[118] = 0xa45

BTHSD16K: extend2[117] = 0x7c ; extend2[118] = 0x66

External NXP9890 IIS configuration:

extend2[117] = 0x7c

extend2[118] = 0x67

NOTE

NXP9890 is a third party chip.

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5.13 Third party Smart-AMP NV (IIS interface) configuration

Sample_rate = 0x10

NOTE

It is not recommended to switch the sampling rate of NXP9890 (third party) between 8KHz and 16KHz, and a unified sampling rate be 16KHz is recommended.

extend2[117] = 0x7c

extend2[118] = 0x67

reserved_for_new[1]=0x1

- 0: IIS0(VBC，BT)

- 1: IIS1(OPEN IIS)

reserved_for_new[3] = 0x2

- 0: the data of DSP point 3 is sent to ARM to be used as Ref Signal.
- 1: the data of DSP AuxMIC is sent to ARM through VBC Loop to be used as Ref Signal.
- 2: the data of DSP AuxMIC is sent to ARM through Codec or Smart_AMP Loop to VBC IIS TX to be used as Ref Signal.

NOTE

Digital signal processor (DSP Point2) decoded source code is about 8K to16K.

Figure 5-9 Decoded 16k pcm

data-0002-16k-0000.pcm	2018/1/9 17:16	Raw PCM file	424 KB
data-0003-16k-0000.pcm	2018/1/9 17:16	Raw PCM file	424 KB
data-0004-16k-0000.pcm	2018/1/9 17:16	Raw PCM file	424 KB
data-0005-16k-0000.pcm	2018/1/9 17:16	Raw PCM file	424 KB
data-0001-8k-0000.pcm	2018/1/9 17:16	Raw PCM file	212 KB
data-0006-8k-0000.pcm	2018/1/9 17:16	Raw PCM file	212 KB

5.14 Digital MIC NV configuration

Sample_rate = 0x10

AudioStructure\extend2\extend2[117]=0x7c

AudioStructure\extend2\extend2[118]=0x67

AudioStructure\vbc\reserved_for_new\reserved_for_new[1]=0x1

AudioStructure\vbc\reserved_for_new\reserved_for_new[3]=0

AudioStructure\vbc\rese

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